

Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care: A Comprehensive Analysis of Stakeholder Perspectives and Policy Recommendations

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Abstract—The integration of Artificial Intelligence (AI) into clinical care represents a transformative opportunity to enhance healthcare delivery, reduce costs, and improve patient outcomes. This paper synthesizes stakeholder responses to the Department of Health and Human Services (HHS) Request for Information on accelerating AI adoption in clinical care, analyzing barriers, regulatory frameworks, payment policy reforms, and governance structures. Drawing from 15 stakeholder submissions—including professional associations, technology developers, patient advocacy groups, and healthcare providers—and supporting academic literature, we identify critical challenges including data interoperability gaps, fragmented governance, misaligned reimbursement incentives, workforce constraints, and trust deficits among clinicians and patients. The analysis reveals that successful AI adoption requires a multi-faceted approach: modernized payment models that reward prevention and value-based care, clear accountability frameworks for non-device AI, robust evaluation methodologies incorporating continuous post-deployment monitoring, and patient-centered design principles that preserve human connection. We present eight architectural frameworks proposed by stakeholders, including shared capacity data utilities, real-time data refineries, post-deployment algorithmic surveillance (PDAS), evidence-based medicine on FHIR (EBMonFHIR), and clinical governance structures with named accountability. The paper further examines the future of work implications, including job displacement concerns, emerging roles (AI Clinical Governance Officer, Clinical AI Specialist), educational requirements across career stages, and human-AI collaboration models. Policy recommendations span regulatory, reimbursement, and research domains, emphasizing the need for shared infrastructure, standardized evaluation frameworks, and governance mechanisms that enable innovation while protecting patient safety. This integrated approach provides a roadmap for federal investment and policy development to create an ecosystem where AI can safely and equitably augment clinical care.

Index Terms—Artificial Intelligence, Clinical Care, Healthcare Policy, Regulatory Framework, AI Governance, Digital Health, Interoperability, Patient Safety

I. INTRODUCTION

The rapid evolution of artificial intelligence (AI) technologies presents unprecedented opportunities to transform clinical care delivery, reduce administrative burden, and improve patient

outcomes. As noted in recent policy discussions, AI-enabled tools have the potential to enhance productivity, reduce burden, lower health care costs, and improve health outcomes for patients, caregivers, and communities [1]. However, realizing this potential requires addressing systemic barriers that currently impede adoption across diverse healthcare settings.

The literature demonstrates that AI applications in healthcare span a continuum from administrative workflow optimization to clinical decision support and autonomous diagnostic systems [2]. Generative AI and large language models have introduced new capabilities and risks, particularly in mental health applications where chatbots may operate in regulatory gray areas [3], [4]. The World Health Organization has emphasized the need for ethics and governance guidance for large multi-modal models in healthcare [5], while recent regulatory developments in the EU and U.S. reflect growing recognition of AI's unique oversight requirements [6], [7].

This paper examines the current landscape of AI in clinical care, analyzes barriers to adoption, and presents policy recommendations grounded in academic literature and stakeholder perspectives.

II. PROPOSED ARCHITECTURAL FRAMEWORKS FOR CLINICAL AI INTEGRATION

Drawing from stakeholder submissions to the HHS RFI, this section synthesizes proposed architectural frameworks for clinical AI integration. These proposals address critical infrastructure needs including data interoperability, governance mechanisms, and continuous monitoring capabilities, with each framework linked to relevant academic literature and established standards.

A. Shared Capacity Data Utility Model

Van Pelt & Company proposes a "shared capacity data utility" model designed as vendor-agnostic, public-private infrastructure that leverages existing systems and non-PHI operational data to provide continuously available views of healthcare system

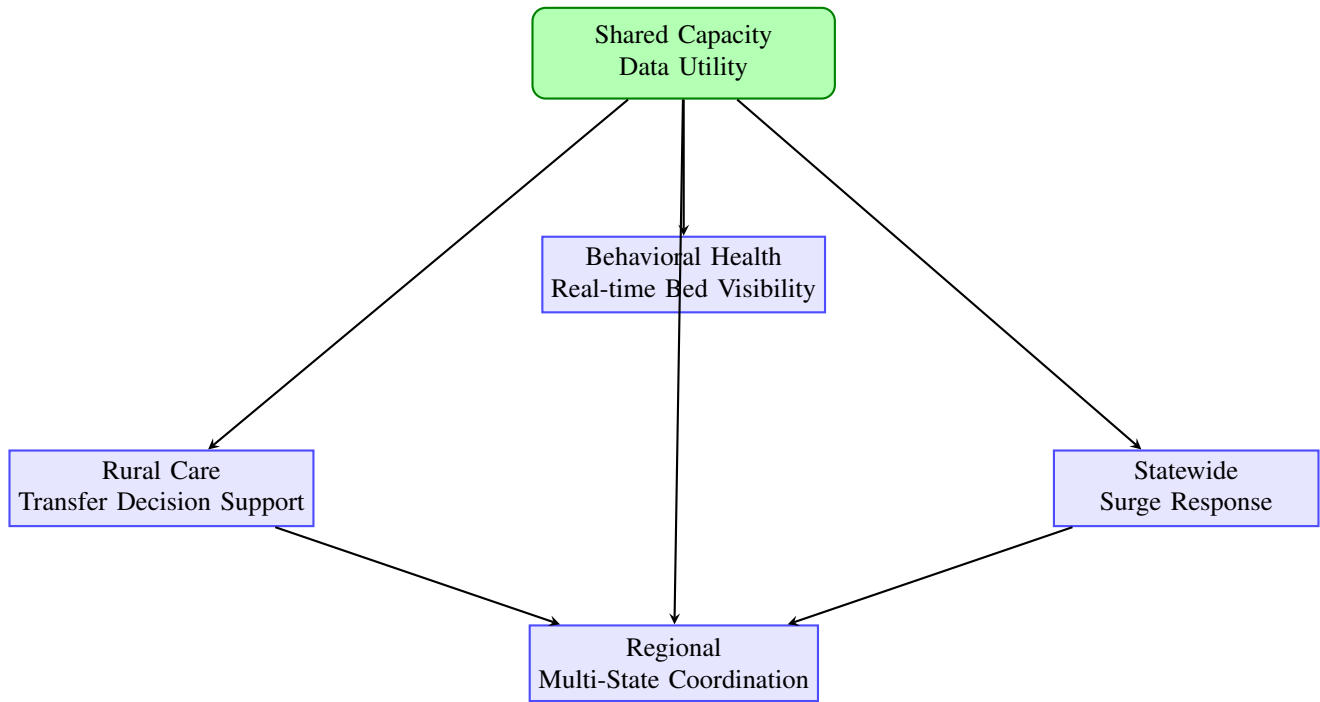


Fig. 1: Shared Capacity Data Utility Model operating across behavioral health, rural care, statewide coordination, and regional multi-state settings. This vendor-agnostic infrastructure leverages existing systems and non-PHI operational data to provide continuously available views of healthcare system capacity [8].

capacity. This framework addresses the foundational gap where AI tools are frequently deployed as point solutions optimized for narrow workflows but blind to system-level constraints such as staffing shortages, bed availability, and regional surge conditions [8].

The utility model operates across multiple care settings:

- **Behavioral health:** Real-time visibility into bed availability and staffing constraints enables faster placement decisions and reduces emergency department boarding. This aligns with research on AI chatbots for mental health that emphasizes the need for contextual awareness in clinical decision support [9], [10].
- **Rural care:** Supports precise decision-making about when patients can be safely managed locally versus requiring transfer to higher-acuity care, addressing equity concerns documented in the literature on healthcare AI access disparities [11].
- **Statewide coordination:** Provides continuous situational awareness for preparedness, surge response, and alignment with federal programs, consistent with WHO guidance on health system resilience [5].
- **Regional multi-state coordination:** Supports interoperability across jurisdictional boundaries for disaster response and medical surge scenarios, as outlined in federal interoperability frameworks [12].

B. Data Refinery Architecture for AI Readiness

Interstella's Lynqsys platform architecture emphasizes real-time data integration and AI readiness through a "Data Refinery

as a Service" (DRaaS) model [13]. The proposed architecture addresses core barriers including delayed data availability, inconsistent data quality, and limited support for unstructured data. Key components include:

- **Real-time data streams:** Moving from batch processes with lag times of days or weeks to near-real-time streaming architectures that enable immediate AI analysis. This addresses data quality concerns identified in the literature where "AI systems do not fail because of algorithms; they fail because of data" [14].
- **Inline data governance:** Privacy protections and quality checks applied at the point of data creation rather than as post-hoc audits, consistent with HIPAA requirements and recent OCR guidance [15].
- **AI-ready data repositories:** Shared datasets and testbeds for AI training and evaluation, supporting the development of representative, high-quality data for model validation as emphasized in the Framework for AI Tool Assessment in Mental Health [16].
- **Support for unstructured data:** Integration of clinical notes, images, and behavioral health data, addressing the limitation that many current systems handle structured data but struggle with unstructured information critical for AI performance [2].

C. Continuous Monitoring and Post-Deployment Surveillance Framework

Dr. Akshaya Bhagavathula proposes a "Post-Deployment Algorithmic Surveillance" (PDAS) framework for governing

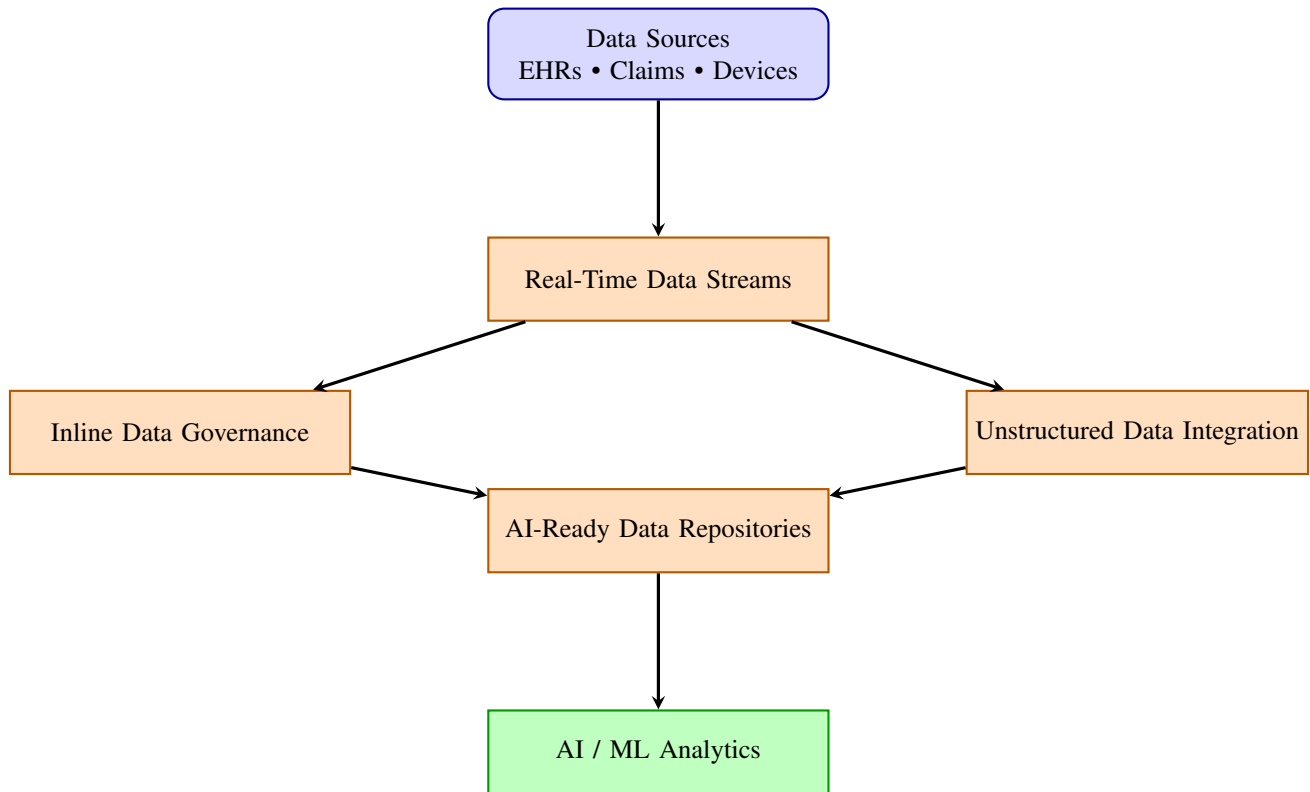


Fig. 2: Interstella's Lynqsys platform architecture emphasizing real-time data integration, inline governance, and AI-ready repositories through a "Data Refinery as a Service" (DRaaS) model [13].

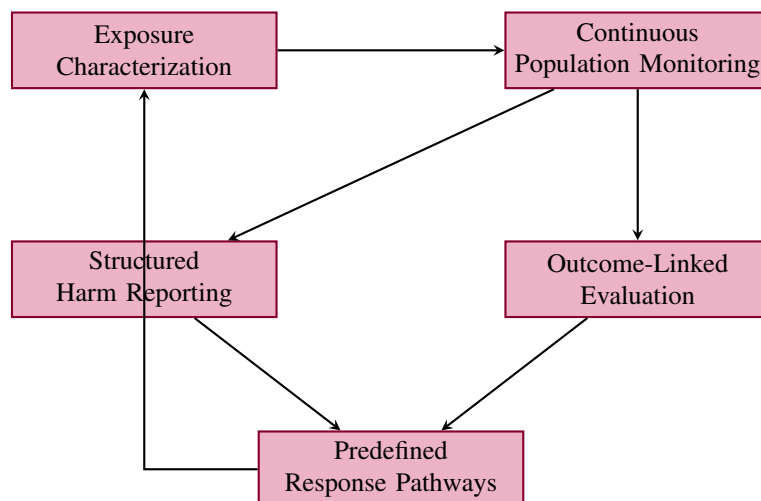


Fig. 3: Post-Deployment Algorithmic Surveillance (PDAS) framework for governing clinical AI as population-level interventions, incorporating exposure characterization, continuous monitoring, outcome evaluation, harm reporting, and response pathways [17].

clinical AI as population-level interventions rather than discrete products [17]. This framework recognizes that once introduced into a health system, AI systems reshape decisions, attention allocation, workflows, and care delivery across patient populations.

The PDAS framework includes five core elements:

- 1) **Exposure characterization:** Systematic measurement of who is exposed to AI-mediated decisions, how often, and in what contexts, establishing foundational data for causal assessment.
- 2) **Continuous population-level performance monitoring:** Longitudinal evaluation of calibration, discrimination, and error patterns with mandatory stratification by key subgroups, treating performance drift as a safety signal rather than technical footnote.
- 3) **Outcome-linked evaluation:** Assessment of whether AI use improves downstream clinical outcomes, access, or utilization patterns, moving beyond surrogate metrics.
- 4) **Structured reporting of algorithmic harms:** Mechanisms to capture delayed care, silent misclassification, workflow distortions, and disparity amplification that rarely trigger traditional adverse event reports.
- 5) **Predefined regulatory response pathways:** Clear protocols for recalibration, restriction, suspension, or withdrawal when surveillance identifies concerns, with documented enforcement actions.

D. Evidence-Based Medicine on FHIR (EBMonFHIR) Architecture

SKAF proposes an interoperable framework for representing scientific knowledge to support AI-generated clinical recommendations [18]. Building on HL7 FHIR standards, this architecture enables both human-interpretable and machine-interpretable representation of evidence supporting AI recommendations, addressing FDA guidance requiring that healthcare professionals be able to "independently review the basis for such recommendations."

The EBMonFHIR Implementation Guide defines approximately 100 profiles, extensions, and value sets for representing:

- Citations for identification, location, classification, and attribution of knowledge artifacts
- Research study design components including eligibility criteria and endpoint analysis plans
- Research results including statistic findings, variable definitions, and evidence certainty
- Aggregation and synthesis of research results across multiple studies
- Judgments regarding evidence syntheses and contextual factors related to recommendations
- Recommendations themselves with supporting evidence

A key component is the SummaryOfNetEffect Profile, which combines data elements for clinical decisions including population specification, intervention and comparator definitions, outcome considerations with risk estimates and effect changes, desirability and relative importance weighting, and

net effect estimates combining contributions [18]. This enables transparent, auditable AI recommendations that clinicians can independently evaluate, addressing concerns about generative AI producing contextually incorrect rationales to support predetermined answers.

E. Reference Architecture for Non-Medical Device AI

BlueHalo LLC proposes a reference architecture for non-medical device AI that describes how evaluation, monitoring, interoperability, and governance components integrate across the AI lifecycle [19]. This architecture addresses the reality that many AI tools operate outside traditional regulatory pathways yet significantly influence clinical workflows.

The proposed architecture includes:

- **Evaluation layer:** Pre-deployment testing in controlled environments combined with continuous post-deployment assessment across diverse populations and settings, addressing the limitation that "validation studies answer whether an algorithm can perform under specified conditions, but they do not answer whether the system remains safe, effective, or equitable once it reshapes real-world clinical behavior" [17].
- **Monitoring layer:** Continuous drift detection, performance tracking, and safety signal surveillance with automated flagging of deviations from expected behavior, enabling early identification of degradation before adverse outcomes occur.
- **Interoperability layer:** Extended beyond data exchange to include metadata, data provenance, evaluation artifacts, and governance policies, providing context necessary for interpretation and accountability.
- **Governance layer:** Clear role definitions, oversight mechanisms, and escalation paths that reduce ambiguity and enable confident deployment in regulated environments.

F. Clinical Governance and Accountability Framework

Sancian LLC proposes an operational framework for AI governance emphasizing named accountability structures and readiness assessment before deployment [20]. Key elements include:

- **Named AI Executive Owner model:** Single accountable leader with cross-functional authority to own AI outcomes, preventing the diffusion of responsibility that occurs when multiple stakeholders hold effective veto power.
- **Cross-functional AI governance councils:** Formalized structures with defined decision rights including clinical, technical, compliance, and patient safety leadership.
- **AI ReadyCheck™ assessment:** Organizational readiness baseline evaluating data governance, workflow optimization, change management capacity, and leadership alignment before technology procurement.
- **AI-EQUITYClear™ assessment:** Equity impact evaluation before deployment to identify potential disparities across demographic subgroups.

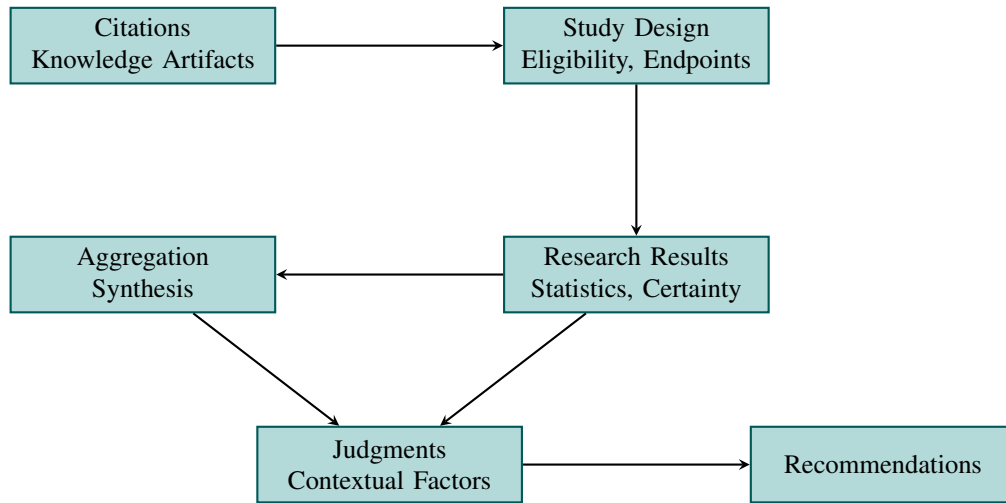


Fig. 4: EBMonFHIR Implementation Guide defining approximately 100 profiles, extensions, and value sets for representing scientific knowledge in human-interpretable and machine-interpretable formats [18].

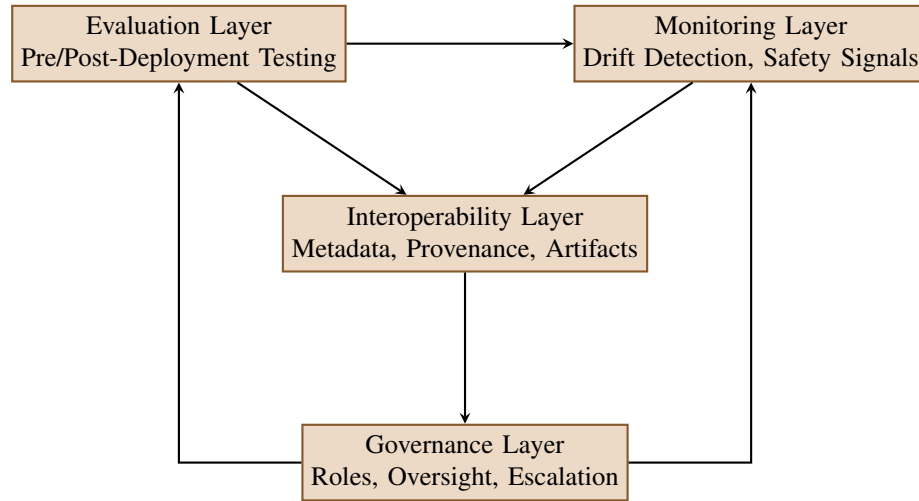


Fig. 5: BlueHalo's reference architecture for non-medical device AI integrating evaluation, monitoring, interoperability, and governance layers across the AI lifecycle [19].

- **Decision clarity before tooling:** Governance infrastructure established before product evaluation, ensuring organizations can responsibly manage AI systems once deployed.
- Validation requirements for AI models across diverse clinical settings
- Continuous quality monitoring including daily quality control, proficiency testing, and trend analysis

G. Laboratory Data Harmonization Infrastructure

The Association for Diagnostics & Laboratory Medicine proposes infrastructure for harmonizing clinical laboratory test results to enable AI interoperability across institutions [21]. This addresses the critical barrier that different testing platforms produce varying numeric results for the same analyte, degrading model performance when data are pooled.

The proposed infrastructure includes:

- Standardized calibration protocols across assay systems
- Metadata inclusion for test methods and validation parameters in AI models
- Uniform guidelines for sample collection and reporting

H. Dental AI Standards and Validation Framework

The Dental Standards Institute proposes a national dental AI trust framework addressing the unique challenges of AI adoption in dentistry, where diagnostic coding is inconsistently used, imaging workflows vary widely, clinical documentation remains largely unstructured, and dental and medical records remain disconnected [22].

The proposed framework includes:

- Standardized validation datasets across multiple imaging devices and clinical settings
- Third-party testing and certification mechanisms

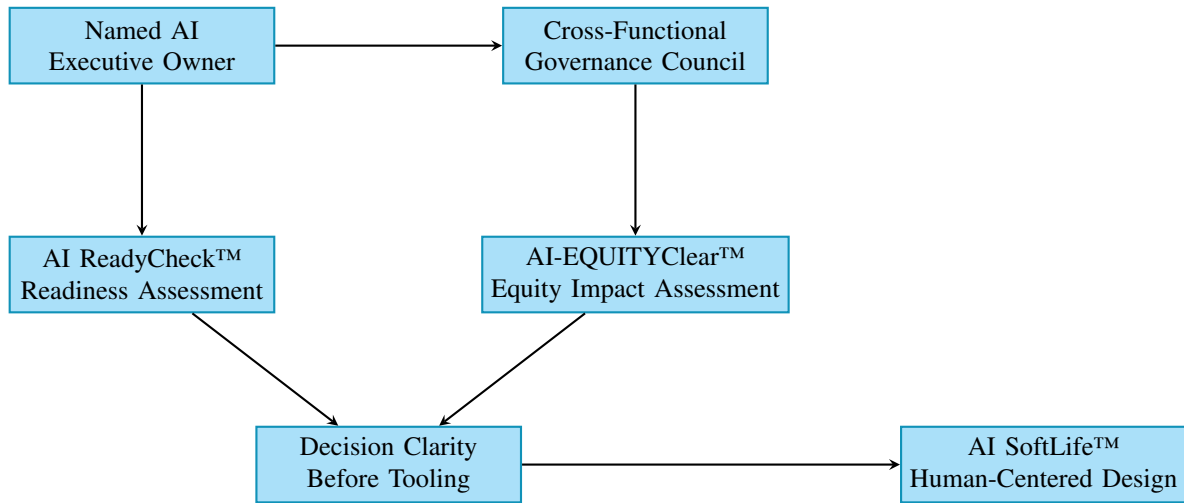


Fig. 6: Sancian LLC's operational framework for AI governance emphasizing named accountability structures, readiness assessment, equity evaluation, and human-centered design principles [20].

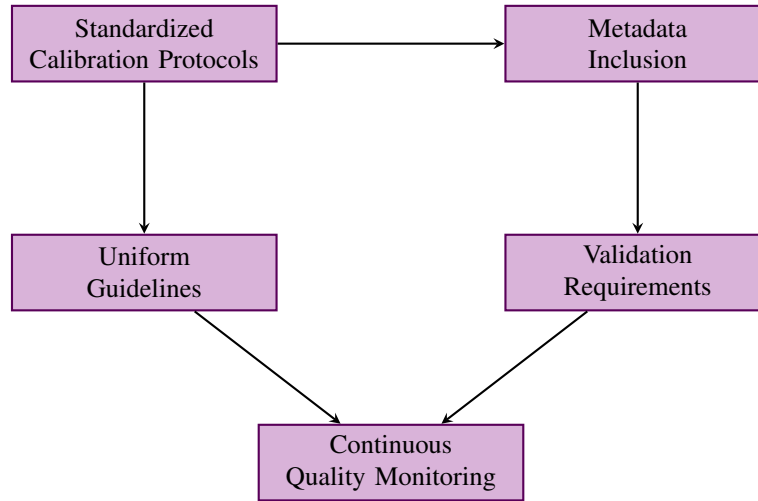


Fig. 7: ADLM's proposed infrastructure for harmonizing clinical laboratory test results to enable AI interoperability across institutions, addressing variability across testing platforms and methods [21].

- Minimum transparency requirements for dental AI vendors
- Standardized performance reporting formats
- Audit log requirements for AI recommendations in clinical workflows
- Guidance for post-deployment monitoring and model drift surveillance

This infrastructure supports the modernization of diagnosis capture in dentistry, enabling structured diagnostic labeling essential for AI training and validation. The framework emphasizes that "self-validation is not validation" and promotes independent benchmarking to ensure patient safety and clinical accountability.

I. Integration of Proposed Architectures

The proposed architectural frameworks are complementary and can be integrated to form a comprehensive AI governance and implementation ecosystem. Figure 9 illustrates how these

frameworks interconnect to address the full spectrum of clinical AI adoption challenges.

The integrated ecosystem demonstrates that successful clinical AI adoption requires coordinated investment across multiple dimensions:

- **Data infrastructure:** Shared capacity utilities, data refineries, laboratory harmonization, and domain-specific frameworks (dental) provide the foundation for AI-ready data.
- **Standards:** EBMonFHIR and similar interoperability standards enable consistent representation of clinical knowledge and AI recommendations.
- **Monitoring:** PDAS frameworks ensure continuous evaluation of AI performance and safety in real-world settings.
- **Governance:** Reference architectures and clinical governance frameworks establish accountability structures and

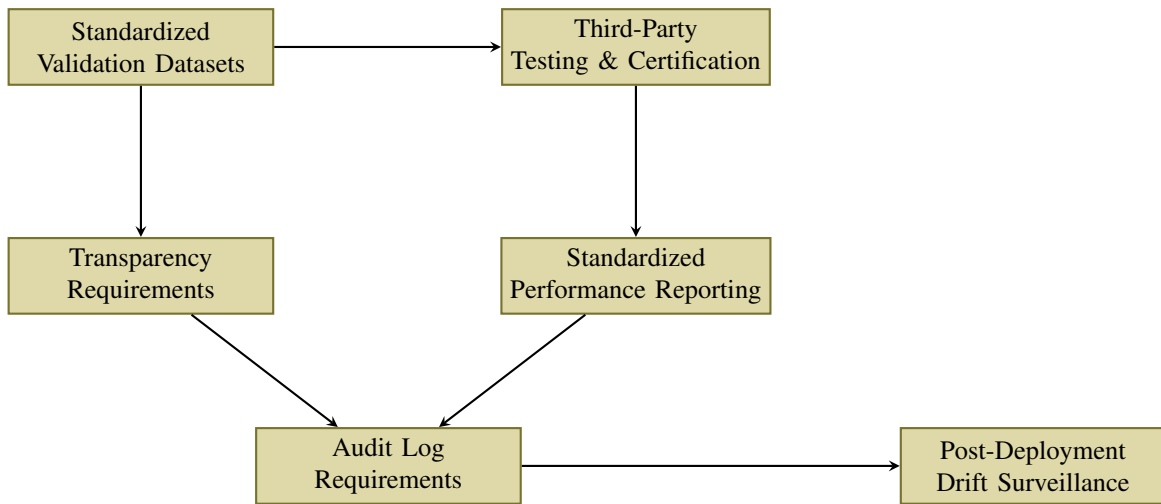


Fig. 8: DSI's national dental AI trust framework addressing unique challenges in dentistry including inconsistent diagnostic coding, imaging workflow variability, and disconnected medical records [22].

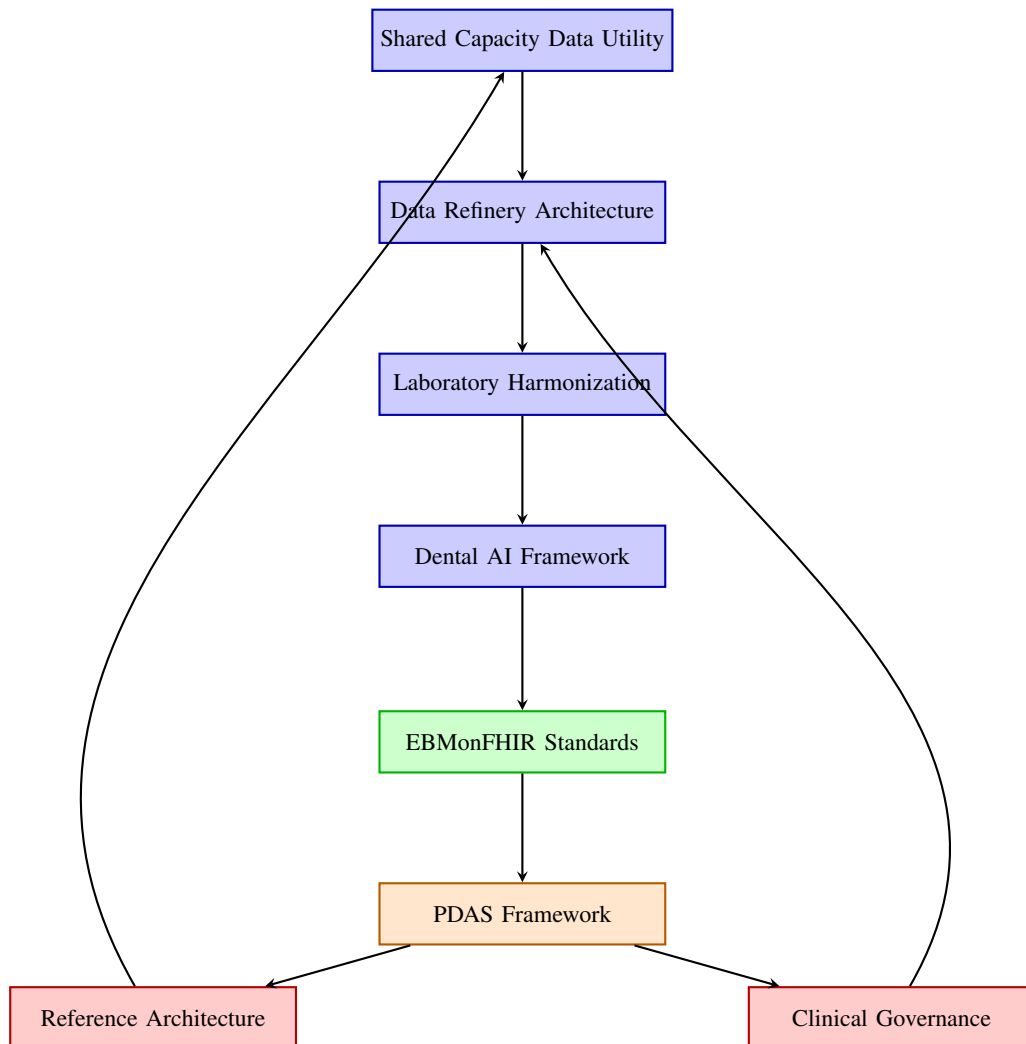


Fig. 9: Integration of proposed architectural frameworks showing relationships between data infrastructure, standards, monitoring, and governance layers.

oversight mechanisms.

This integrated approach addresses the systemic barriers identified in stakeholder submissions while providing a roadmap for federal investment and policy development. By treating AI as care infrastructure requiring coordinated governance rather than isolated technology deployment, HHS can accelerate adoption while maintaining patient safety and public trust.

III. TECHNICAL ARCHITECTURE REVIEW OF PROPOSED AI FRAMEWORKS IN CLINICAL CARE

This section provides a comprehensive technical review of the architectural frameworks proposed by stakeholders in response to the HHS RFI. We analyze the technical components, data flows, integration patterns, and AI/ML model architectures that underpin these proposals, with particular attention to agentic AI systems, generative models, and the infrastructure required for their safe deployment in clinical settings.

A. Taxonomy of AI Models and Agents in Clinical Architectures

The proposed architectures employ a diverse range of AI models and agent types, each suited to specific clinical tasks. Table 1 presents a taxonomy of AI models referenced in stakeholder submissions, mapping them to architectural components and clinical applications.

B. Agentic AI Architectures for Clinical Workflow Automation

Agentic AI systems—autonomous AI agents capable of perceiving their environment, making decisions, and taking actions—represent a significant advancement in clinical workflow automation. Figure 10 illustrates a generalized agentic AI architecture synthesized from multiple stakeholder proposals.

The agentic architecture comprises several key components:

- **Data Integration Layer:** Aggregates structured and unstructured data from EHRs, claims systems, IoT devices, and patient-reported sources. Interstella’s DRaaS model emphasizes real-time streaming and inline governance at this layer [13].
- **Agent Orchestrator:** Coordinates task allocation among specialized agents, manages agent communication, and maintains system state. The orchestrator implements safety protocols and escalation pathways to human clinicians.
- **Specialized Agent Pool:** Includes scheduling agents for appointment optimization, triage agents for initial patient assessment, care coordination agents for longitudinal management, predictive analytics agents for risk stratification, and patient outreach agents for engagement [27], [28].
- **Human-in-the-Loop Interface:** Ensures that autonomous actions are reviewed by clinicians when they exceed predefined confidence thresholds or involve high-risk decisions, as emphasized by Sancian’s AI SoftLife™ philosophy [20].

C. Generative AI Integration Patterns

Generative AI models, particularly large language models (LLMs), are integrated into clinical architectures through several patterns. Figure 11 illustrates the primary integration patterns identified in stakeholder submissions.

1) *Retrieval-Augmented Generation (RAG) Pattern:* The RAG pattern combines LLMs with external knowledge bases to ground responses in verified clinical evidence. SKAF’s EBMonFHIR architecture exemplifies this pattern by integrating FHIR-based clinical data with LLM-generated recommendations [18]. Key technical components include:

- **Vector Databases:** Store embeddings of clinical guidelines, research literature, and institutional protocols for semantic similarity search.
- **Retrieval Mechanisms:** Implement hybrid search combining keyword matching, semantic similarity, and knowledge graph traversal.
- **Context Integration:** Retrieved evidence is injected into LLM prompts to ensure responses are evidence-based and citeable.

2) *Fine-Tuning for Domain Adaptation:* Fine-tuning adapts base LLMs to clinical domains using specialized datasets. Campellone et al. demonstrated that fine-tuned models achieved 98% accuracy in detecting and responding empathetically to patient statements, compared to 69% for rules-based systems [4]. Technical considerations include:

- **Training Data:** Curated clinical conversations, annotated with safety markers and empathy scores.
- **Parameter-Efficient Fine-Tuning:** LoRA and adapter techniques enable domain adaptation with minimal computational overhead.
- **Validation Protocols:** Rigorous testing across demographic subgroups to detect bias and ensure equitable performance [11].

3) *Guardrail Pattern for Safe Deployment:* Guardrails are critical for safe deployment of generative AI in clinical settings. Ellipsis Health’s architecture incorporates multiple guardrail layers [23]:

- 1) **Input Guardrails:** Detect harmful inputs, off-topic queries, and crisis indicators (suicidal ideation, self-harm).
- 2) **Model Guardrails:** Constrain output generation through controlled decoding, safety classifiers, and refusal mechanisms.
- 3) **Output Guardrails:** Validate generated content against clinical guidelines, toxicity filters, and fact-checking systems.
- 4) **Escalation Guardrails:** Trigger human clinician involvement when inputs or outputs exceed safety thresholds.

D. Federated Learning Architectures for Privacy-Preserving AI

Federated learning enables collaborative model training across institutions without centralizing sensitive patient data. Figure 12 illustrates the federated learning architecture proposed for multi-institutional AI development.

Key technical components of federated learning architectures include:

- **Local Training:** Each institution trains models on local data, generating gradient updates or model weights.

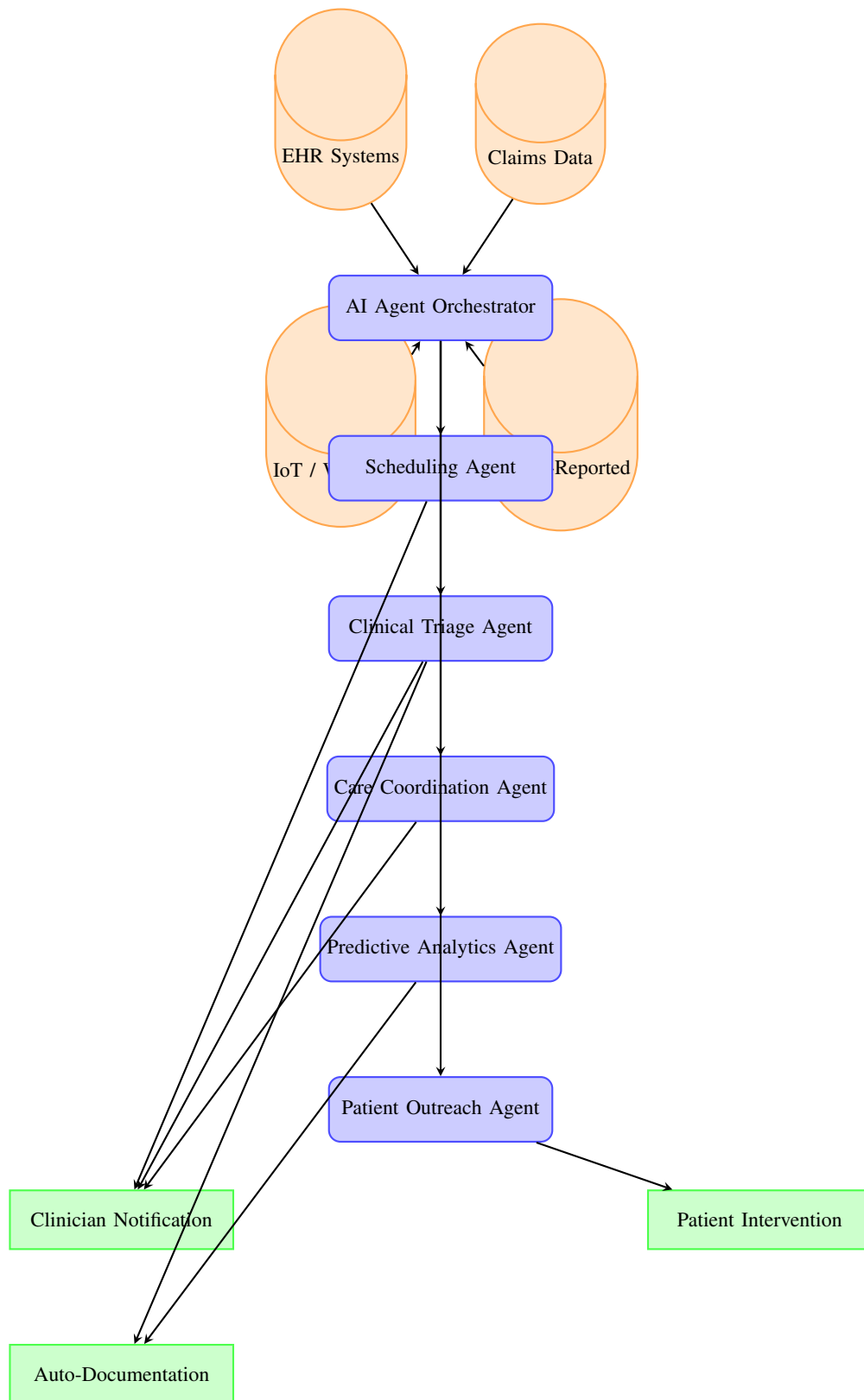


Fig. 10: Agentic AI Architecture for Clinical Workflow Automation showing the orchestration layer, specialized agent pool, and task execution components. This architecture enables autonomous decision-making and action execution while maintaining human oversight through clinician notification pathways [14], [20].

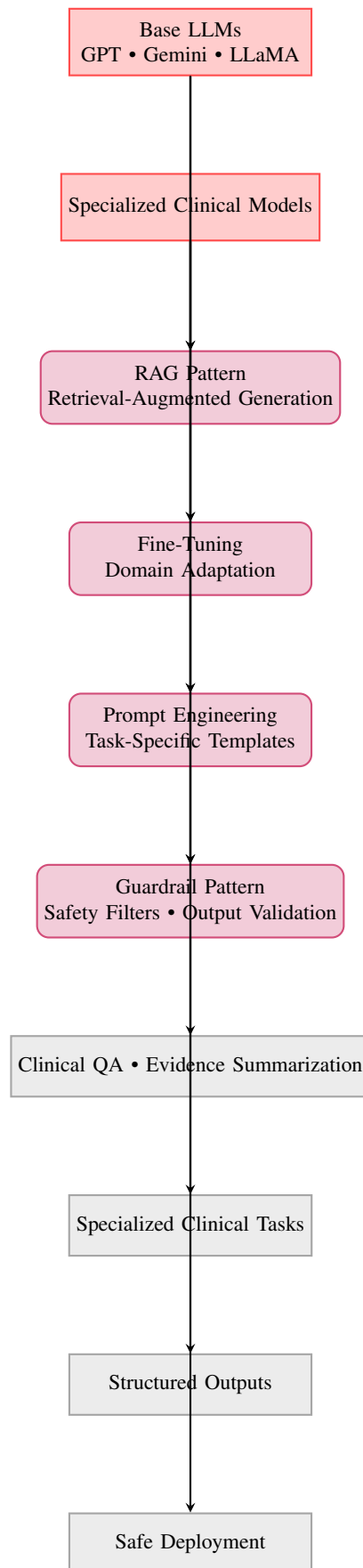


Fig. 11: Generative AI integration patterns in clinical architectures showing four primary approaches: Retrieval-Augmented Generation (RAG) for evidence-based responses, fine-tuning for domain adaptation, prompt engineering for structured task outputs, and guardrail patterns for safety validation [4], [23].

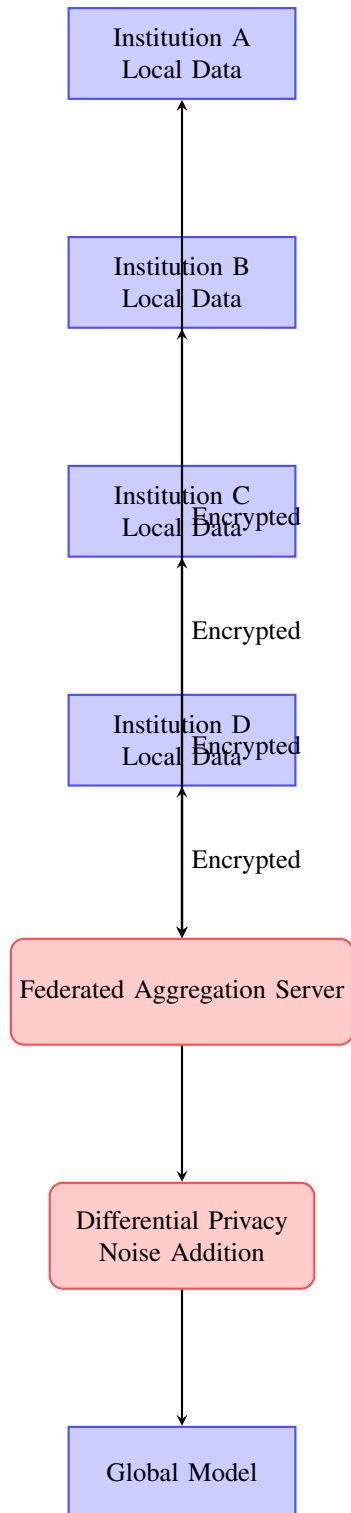


Fig. 12: Federated Learning Architecture for Privacy-Preserving Multi-Institutional AI Development. Local institutions train models on their data and share only encrypted model updates with the central aggregator, which combines updates using secure aggregation protocols before distributing the improved global model [26], [14].

TABLE I: Taxonomy of AI Models and Agents in Clinical Architectures

Model/Agent Type	Architectural Component	Clinical Application	Reference Submissions
Generative AI (LLMs)	Conversational Interfaces, Clinical Documentation	Mental health chatbots, ambient listening, clinical note generation	Ellipsis Health [23], AORN [24]
Agentic AI Systems	Autonomous Decision Support, Workflow Orchestration	Care coordination, prior authorization, patient outreach	Joshi Framework [14], Sancian LLC [20]
Predictive ML Models	Risk Stratification, Early Warning Systems	Sepsis prediction, readmission risk, deterioration detection	Bhagavathula PDAS [17]
Computer Vision Models	Imaging Analysis, Diagnostic Support	Radiology, pathology, dental imaging	DSI [22], IAC [25]
NLP Models	Unstructured Data Processing	Clinical notes, social determinants, patient-reported outcomes	Interstella [13]
Reinforcement Learning	Adaptive Treatment Optimization	Chronic disease management, personalized care pathways	BlueHalo [19]
Federated Learning Models	Privacy-Preserving Distributed Training	Multi-institutional model development	Nandagopal [26]
Knowledge Graph Models	Clinical Decision Support, Evidence Synthesis	EBMonFHIR, guideline-based recommendations	SKAF [18]

- **Secure Aggregation:** Cryptographic protocols (homomorphic encryption, secure multi-party computation) combine updates without revealing individual contributions.
- **Differential Privacy:** Noise injection ensures that aggregated models cannot be reverse-engineered to extract patient-level information.
- **Federated Averaging:** The central server computes weighted averages of local updates to produce improved global models.

E. Data Pipeline Architectures for Real-Time AI

Real-time AI requires sophisticated data pipeline architectures that minimize latency while maintaining data quality and governance. Figure ?? illustrates the data pipeline architecture proposed by Interstella and others.

F. Knowledge Graph and Semantic Reasoning Architectures

SKAF's EBMonFHIR architecture leverages knowledge graphs and semantic reasoning to enable evidence-based AI recommendations. Figure 14 illustrates the knowledge graph architecture.

G. Continuous Monitoring and Drift Detection Architectures

Bhagavathula's Post-Deployment Algorithmic Surveillance (PDAS) framework incorporates continuous monitoring and drift detection. Figure 15 illustrates the technical architecture for real-time AI monitoring.

H. Color-Coded Reference Architecture Summary

Figure ?? presents a color-coded summary of the integrated architectural layers, mapping technical components to functional domains.

I. Technical Requirements and Implementation Considerations

Based on the architectural analysis, Table II summarizes the technical requirements for implementing these frameworks.

J. Summary

The technical architectures proposed by stakeholders represent a comprehensive ecosystem for clinical AI deployment. Key themes include:

- **Modularity:** Layered architectures enable independent evolution of data infrastructure, models, agents, and governance.
- **Safety-by-Design:** Guardrails, monitoring, and response mechanisms are integrated from the ground up.
- **Interoperability:** FHIR-based standards ensure compatibility across institutions and systems.
- **Scalability:** Cloud-native architectures support deployment from small clinics to large health systems.
- **Privacy-Preservation:** Federated learning and differential privacy enable collaborative development without data sharing.

These technical frameworks provide the foundation for safe, effective, and equitable AI adoption in clinical care, addressing the systemic barriers identified in stakeholder submissions while enabling innovation at scale.

IV. FUTURE OF WORK WITH AI ADOPTION: WORKFORCE TRANSFORMATION, JOB DISPLACEMENT, AND EDUCATIONAL CHALLENGES IN CLINICAL CARE

The integration of artificial intelligence into clinical care represents not merely a technological shift but a fundamental transformation of healthcare work itself. This section examines the multifaceted implications of AI adoption for the healthcare

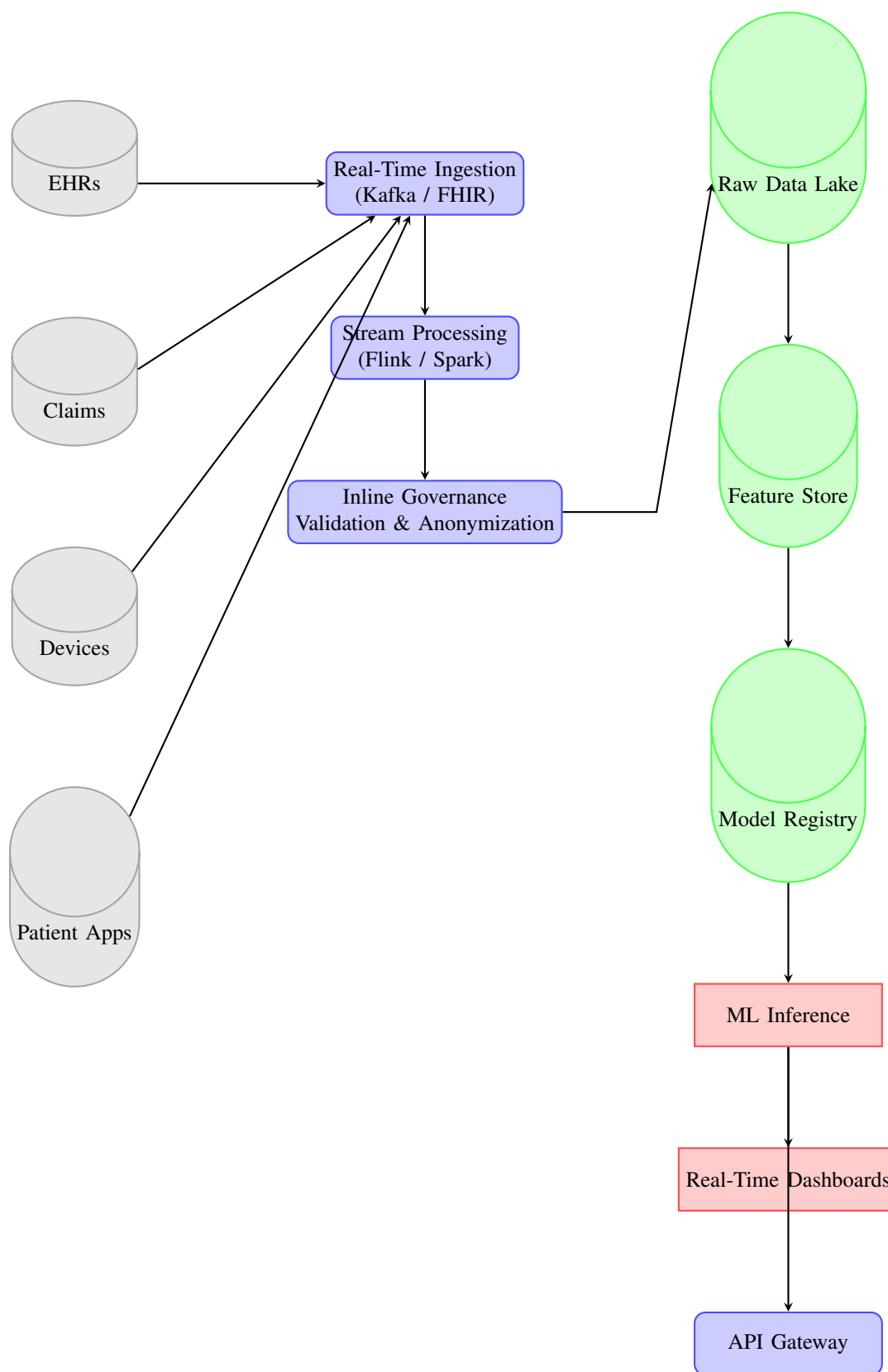


Fig. 13: Two-Column AI Data Pipeline Architecture for Multi-Layer Clinical Workflows. Data sources, ingestion, and stream processing are shown in the left column, while storage, analytics, and serving layers are in the right column, preserving logical flow for narrow multi-column publications [13], [19].

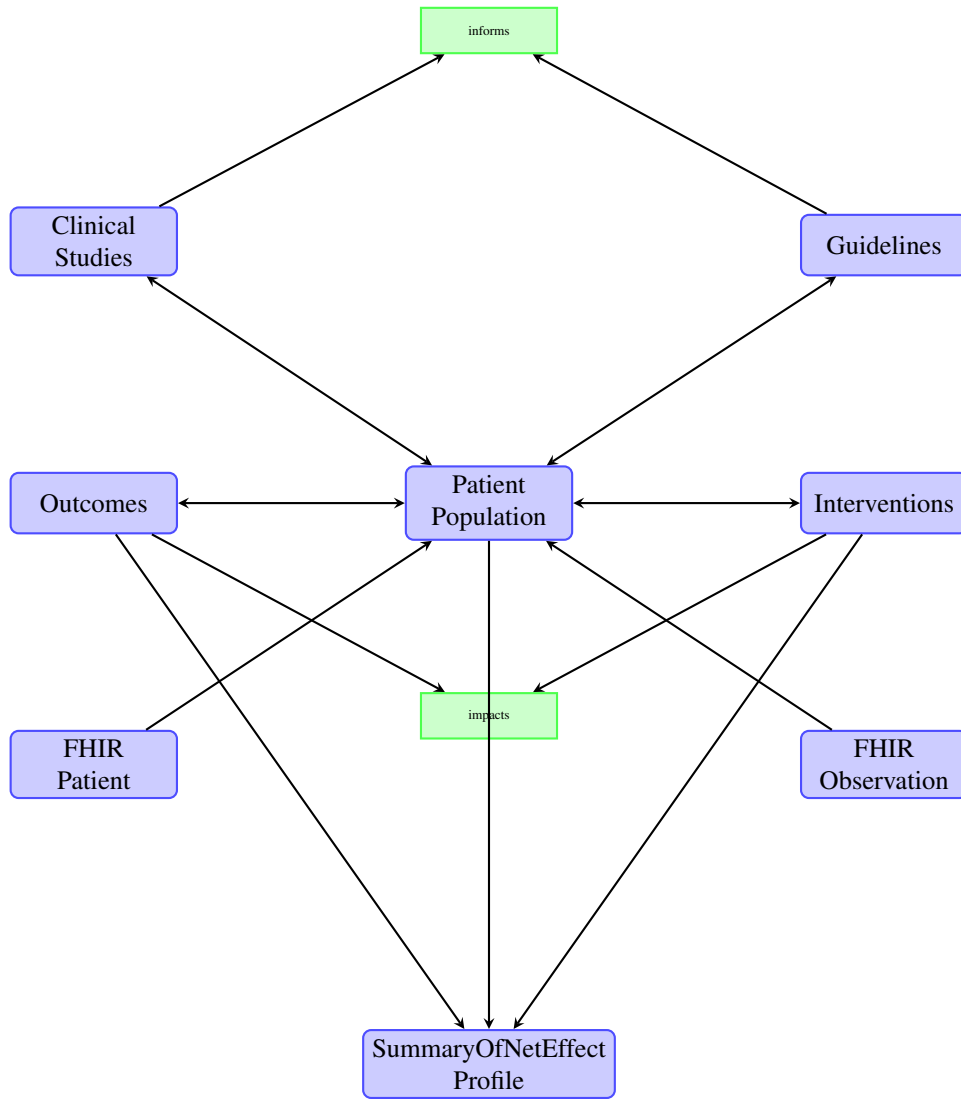


Fig. 14: Knowledge Graph Architecture for Evidence-Based AI Recommendations. The graph integrates clinical studies, guidelines, outcomes, and interventions with FHIR resources, enabling semantic reasoning and evidence synthesis through the SummaryOfNetEffect Profile [18], [29].

workforce, including job displacement concerns, the emergence of new roles, educational requirements, and the critical importance of human-AI collaboration. We analyze stakeholder perspectives on workforce impact and propose frameworks for managing this transition equitably and effectively.

A. The Workforce Impact Spectrum: Automation vs. Augmentation

The debate surrounding AI's impact on healthcare employment often polarizes between fears of widespread job displacement and visions of seamless augmentation. Reality lies on a spectrum, with different clinical roles experiencing varying degrees of transformation. Figure 17 illustrates this spectrum across healthcare professions.

Ellipsis Health emphasizes that "the most effective healthcare AI does not seek to replace clinicians or focus only on

automation" but rather serves as a "force multiplier for human care" that "extends the reach of care teams" [23]. This perspective frames AI as capacity infrastructure that reduces cognitive burden and enables clinicians to focus on higher-value patient interactions.

B. Job Displacement: Realities and Mitigation Strategies

Concerns about job displacement are legitimate but require nuanced analysis. The Association of perioperative Registered Nurses (AORN) takes a firm position: "AI is suitable for augmentation of clinical decision making, but AI cannot and must never be a replacement for a human's clinical judgement. The human expertise, empathy and accountability required for clinical decisionmaking in practice cannot be replicated" [24].

Figure 18 presents a framework for understanding job displacement risk factors in healthcare.

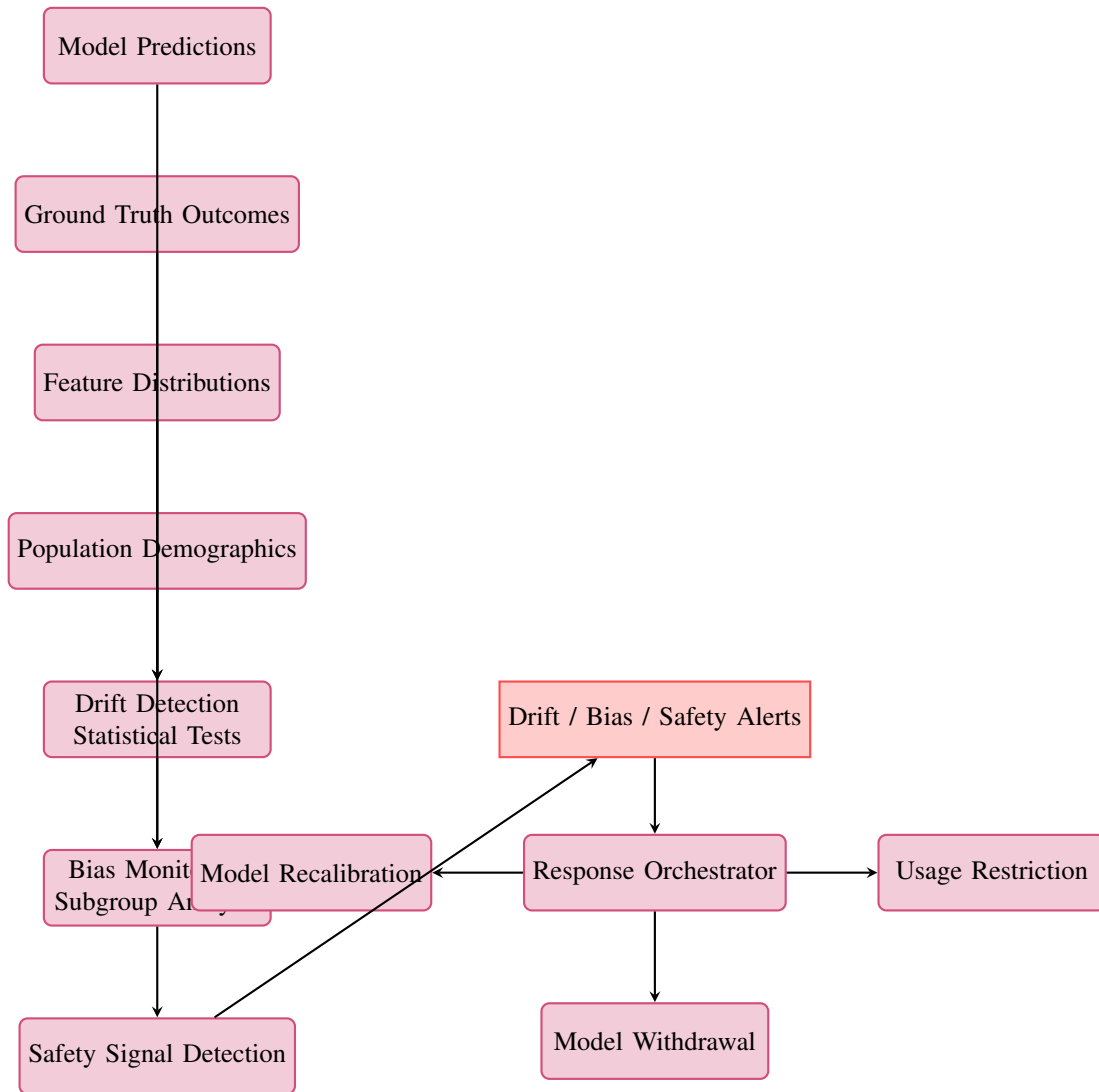


Fig. 15: Continuous Monitoring and Drift Detection Architecture for Clinical AI. The left column contains input and monitoring streams, while the right column contains alerting and response orchestration. This strictly fits within a two-column journal format [17], [30].

1) *Quantitative Impact Estimates:* While precise job displacement estimates vary, several stakeholder submissions provide frameworks for assessment:

- **Administrative roles:** Himmelstein et al. document that administrative work consumes one-sixth of U.S. physicians' working hours [31]. AI automation of prior authorization, documentation, and scheduling could significantly reduce this burden without displacing clinical staff.
- **Diagnostic specialties:** Radiology and pathology face highest automation potential for routine image interpretation, but demand for these specialties may shift toward complex case consultation and oversight of AI systems [29].
- **Nursing and care management:** Ellipsis Health notes that "nurses, care managers, and social workers are already managing unsustainable caseloads" [23]. AI adoption in

this domain is more likely to reduce burnout and enable focus on direct patient care than to eliminate positions.

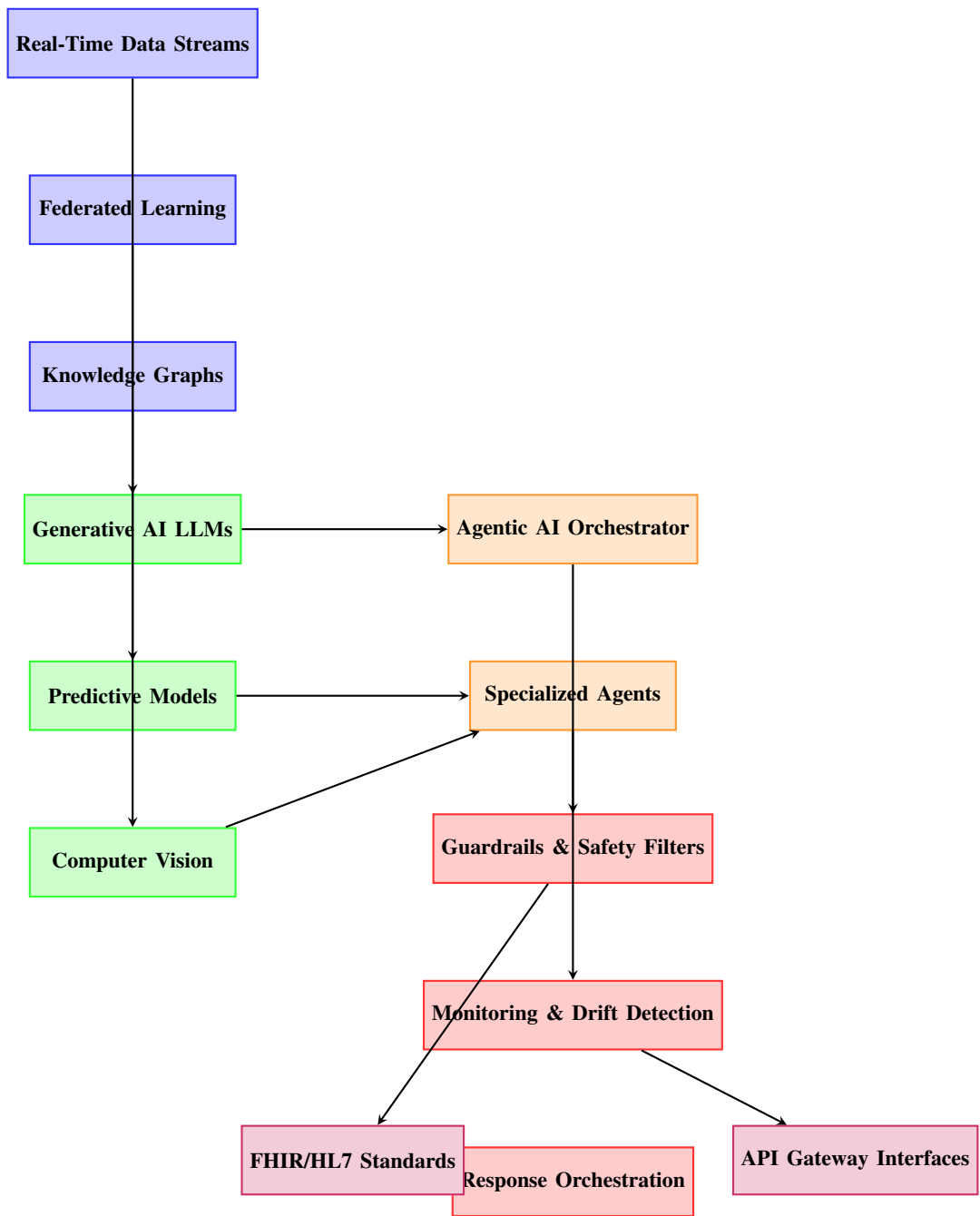
C. Emerging Job Categories and Role Transformation

AI adoption creates new job categories while transforming existing ones. Table III summarizes emerging roles identified in stakeholder submissions.

Sancian LLC's "Named AI Executive Owner" model establishes accountability at the executive level, ensuring that organizations have dedicated leadership for AI governance and workforce transition [20].

D. Educational and Training Requirements

The transformation of healthcare work necessitates corresponding changes in education and training. Figure 19 presents a framework for AI-related education across career stages.



Data Infrastructure



AI Models



Agentic Systems



Governance & Safety

TABLE II: Technical Requirements for Clinical AI Architectures

Component	Technical Requirements	Performance Targets
Data Ingestion	HL7 FHIR APIs, Kafka streams, real-time ETL	Latency < 100ms, Throughput > 10K events/sec
Model Serving	GPU/TPU inference, model versioning, A/B testing	Inference time < 500ms, Availability 99.9%
Agent Orchestration	Workflow engines, state management, inter-agent communication	Task completion < 2s, Scalability to 1000+ concurrent agents
Guardrails	Safety classifiers, output validation, escalation triggers	False positive rate < 1%, Recall > 95% for safety events
Monitoring	Statistical drift detection, bias metrics, safety surveillance	Detection latency < 1 hour, Alert precision > 90%
Federated Learning	Secure aggregation, differential privacy, encrypted communication	Privacy budget $\epsilon < 2$, Model accuracy within 5% of centralized
Knowledge Graphs	Graph databases, semantic reasoning, FHIR integration	Query latency < 200ms, Support for 10M+ triples

AI Impact Spectrum Across Healthcare Roles

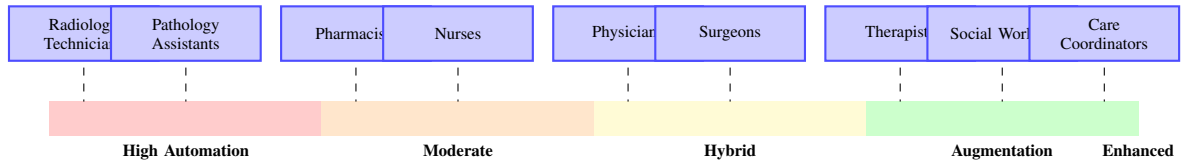


Fig. 17: AI Impact Spectrum Across Healthcare Roles showing the continuum from high automation potential (left) to human augmentation and enhancement (right). Routine pattern recognition roles face higher automation risk, while complex human interaction and judgment roles are more likely to be augmented [11], [23].

TABLE III: Emerging Job Categories in AI-Enhanced Healthcare

Role	Responsibilities	Source
AI Clinical Governance Officer	Oversee AI deployment, ensure safety and equity, manage algorithm lifecycle	Sancian LLC [20]
Clinical AI Specialist	Bridge between clinical practice and AI development, validate model outputs	AORN [24]
AI Quality Assurance Analyst	Monitor model performance, detect drift and bias, conduct audits	Bhagavathula [17]
Human-AI Collaboration Specialist	Optimize workflow integration, train teams on AI tools, manage change	Ellipsis Health [23]
AI Ethics & Equity Officer	Ensure fairness across populations, address algorithmic bias	National MS Society [32]
Data Steward	Manage data quality, governance, and interoperability for AI readiness	Interstella [13]
AI-Enhanced Care Coordinator	Leverage AI insights for patient outreach and care management	Ellipsis Health [23]

Figure 2: Job Displacement Risk Factors in Healthcare AI

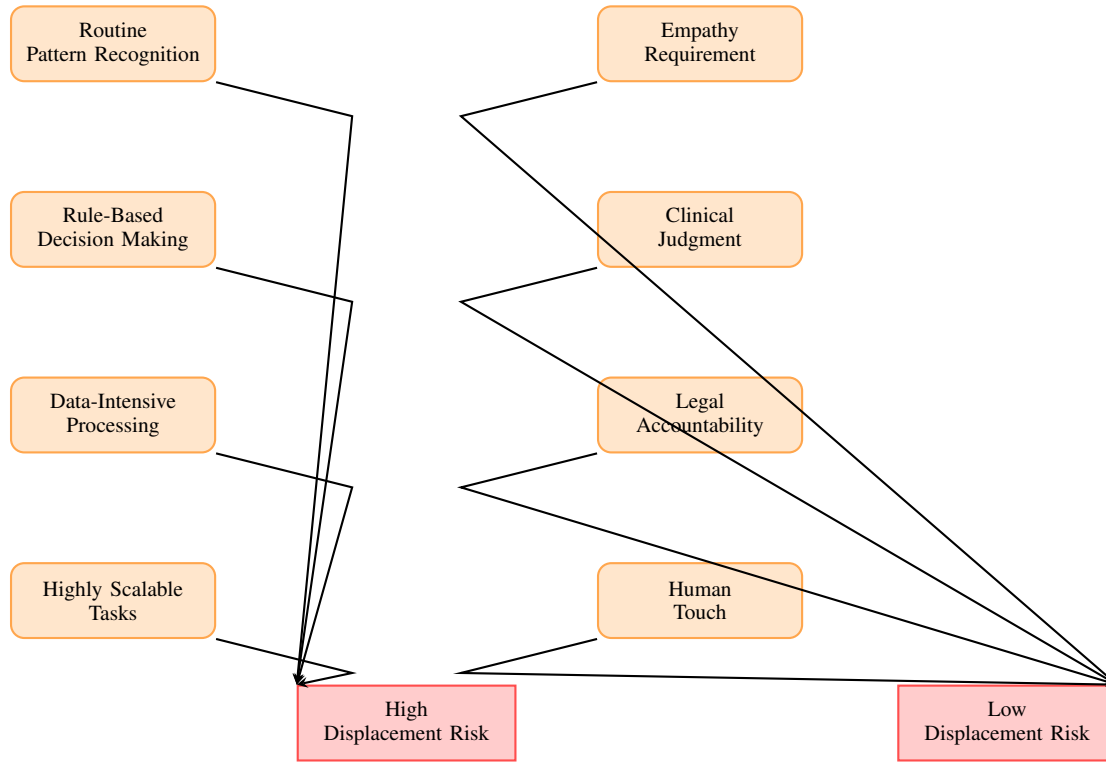


Fig. 18: Job Displacement Risk Factors in Healthcare AI showing the interplay between automation-prone task characteristics (left) and human-essential qualities (right) that determine overall displacement risk. Roles combining high routine content with low empathy requirements face greatest displacement pressure [11], [20].

1) *Core Educational Competencies:* The National Multiple Sclerosis Society emphasizes that "AI education is essential and should be implemented in the academic setting and in the organizations where the AI-enabled technology is used. Competency should be assessed before using any AI-enabled technology" [32].

Key educational competencies identified across stakeholder submissions include:

- 1) **Technical Literacy:** Understanding AI capabilities, limitations, and appropriate use cases.
- 2) **Critical Evaluation:** Assessing AI outputs, recognizing errors, and knowing when to override recommendations.
- 3) **Ethical Awareness:** Understanding bias, fairness, transparency, and accountability in AI systems.
- 4) **Workflow Integration:** Incorporating AI tools into clinical workflows without disruption.
- 5) **Patient Communication:** Explaining AI involvement to patients and addressing concerns.
- 6) **Continuous Learning:** Staying current with evolving AI capabilities and best practices.

E. Labor and Employment Considerations

The integration of AI into clinical care raises significant labor and employment issues that require attention from policymakers, healthcare organizations, and labor representatives.

1) *Scope of Practice and Professional Boundaries:* AORN emphasizes that "clinicians should be full partners in preparing for the future of AI that intersects their respective practice settings" [24]. As AI systems assume tasks traditionally performed by specific professional groups, scope of practice definitions may need revision. Figure 20 illustrates the shifting boundaries.

2) *Liability and Accountability:* Tavory's ethics of care perspective highlights that AI deployment raises "unresolved questions regarding responsibility, liability, and oversight when automated systems fail or when risks go unrecognized" [33]. Key labor-related liability questions include:

- Who is liable when a clinician relies on an AI recommendation that proves harmful?
- How is accountability shared among AI developers, healthcare organizations, and individual clinicians?

Figure 3: AI Education Framework Across Career Stages

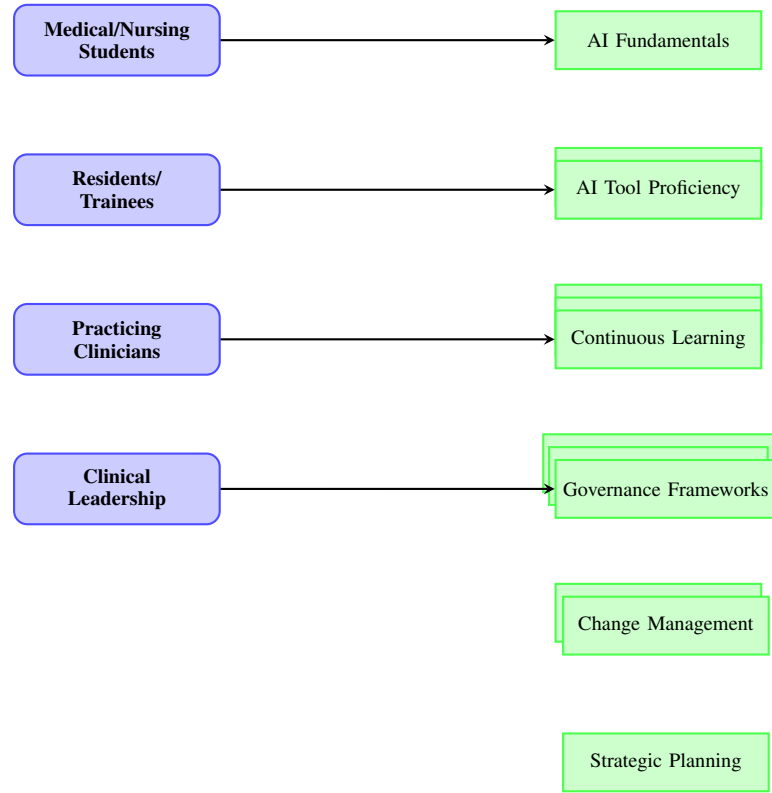


Fig. 19: AI Education Framework Across Career Stages showing progressive educational content from foundational knowledge through clinical integration to governance and leadership. This vertical layout supports columnar publication formats while preserving clarity of learning progression [24], [32].

- What protections exist for clinicians who override AI recommendations that later prove correct?

The Global Liver Institute emphasizes that "clinicians should be transparent with patients about when and how these tools are assisting their care" and that "clear communication and clinician accountability are essential to maintaining trust" [34].

3) *Collective Bargaining and Worker Voice:* As AI transforms healthcare work, mechanisms for worker input into AI deployment decisions become critical. Sancian LLC's governance framework includes "cross-functional AI governance councils" with representation from clinical, technical, compliance, and patient safety leadership [20]. However, formal labor representation in these bodies remains inconsistent.

Anonymous stakeholder comments highlight that "the smaller institutions do not know how to move forward with clinical AI, how to start or what policy they should follow. On the other hand, the larger institutions are overly bureaucratic in making decisions" [35]. This disparity suggests that collective bargaining and worker voice mechanisms may be more accessible in larger, unionized settings.

F. Human-AI Collaboration Models

Successful AI integration depends on effective human-AI collaboration models. Figure 21 presents a taxonomy of collaboration models identified in stakeholder submissions.

Ellipsis Health's "empathy and engagement as core safety features" philosophy emphasizes that "AI tools that patients trust and interact with consistently are more likely to surface early warning signs and less likely to miss critical information" [23]. This perspective frames human-AI collaboration as essential to patient safety, not merely workflow efficiency.

G. Psychological and Emotional Impacts on Workers

The introduction of AI into clinical environments has significant psychological and emotional implications for healthcare workers. Figure 22 illustrates these impacts.

Sancian LLC's "AI SoftLife™ philosophy" explicitly addresses these concerns, designing AI as "capacity infrastructure that reduces cognitive burden, protects clinicians' energy, and sustains humane care delivery" [20]. This human-centered approach recognizes that worker wellbeing is not merely a

Figure 4: Shifting Professional Boundaries with AI Integration

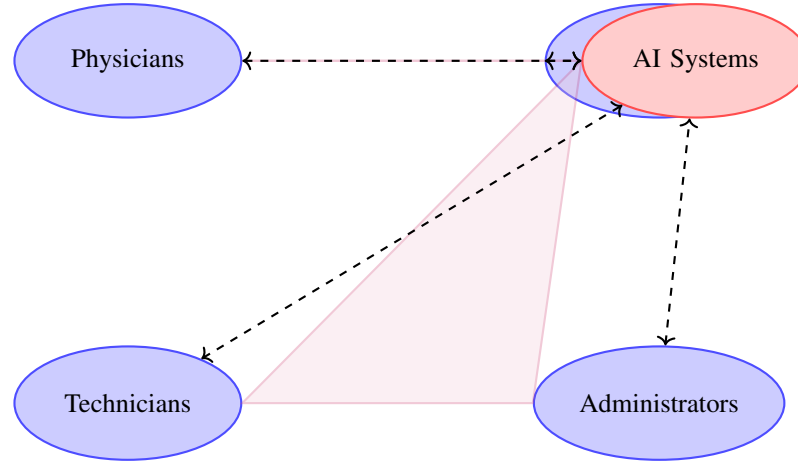


Fig. 20: Shifting Professional Boundaries with AI Integration showing the overlap between traditional professional domains and AI capabilities. As AI systems assume tasks spanning multiple professions, scope of practice definitions must evolve to accommodate new collaborative models [24], [33].

secondary consideration but a prerequisite for successful AI adoption.

H. Recommendations for Workforce Transition

Based on stakeholder submissions, we synthesize recommendations for managing workforce transition:

1) For Healthcare Organizations:

- 1) **Establish governance structures** with worker representation to guide AI deployment decisions [20].
- 2) **Invest in continuous education** and training programs that prepare workers for AI-augmented roles [32], [24].
- 3) **Design AI for augmentation, not replacement**, prioritizing tools that reduce burden and enhance human capabilities [23].
- 4) **Create career pathways** into emerging AI-related roles, offering retraining and upskilling opportunities.
- 5) **Monitor psychological impacts** and provide support for workers experiencing anxiety or resistance.

2) For Policymakers:

- 1) **Fund workforce development programs** specifically targeting AI-related skills in healthcare [36].
- 2) **Update scope of practice regulations** to accommodate AI-augmented care models while protecting professional standards [24].

- 3) **Establish liability frameworks** that fairly allocate responsibility among developers, organizations, and clinicians [33].
- 4) **Support research** on workforce impacts, including longitudinal studies of job transformation and displacement [11].
- 5) **Incentivize inclusive governance** that incorporates worker voice in AI deployment decisions.

3) For Educational Institutions:

- 1) **Integrate AI competencies** across healthcare curricula, from foundational education through continuing professional development [32].
- 2) **Develop interdisciplinary programs** that combine clinical expertise with data science and AI governance.
- 3) **Create simulation environments** where students can safely learn to collaborate with AI systems.
- 4) **Emphasize ethical reasoning** and critical evaluation skills alongside technical proficiency.

I. Summary

The future of work with AI adoption in clinical care presents both challenges and opportunities. While concerns about job displacement are legitimate, the predominant stakeholder perspective emphasizes augmentation over automation, with AI serving to reduce burden, enhance capabilities, and enable

Figure 5: Human-AI Collaboration Models in Clinical Care

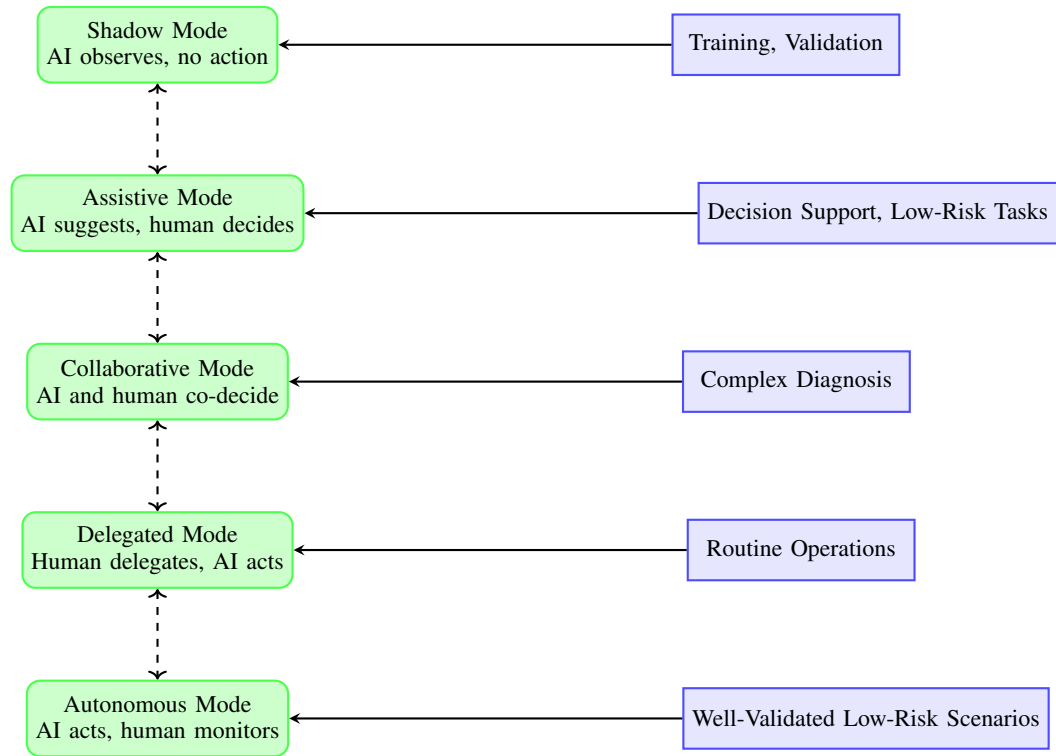


Fig. 21: Human-AI Collaboration Models in Clinical Care showing the progression from AI as observer to AI as autonomous actor. The appropriate mode depends on task complexity, risk, and validation, with most clinical applications operating in assistive or collaborative modes [23], [20].

focus on higher-value patient interactions. Successful workforce transition requires:

- Recognition that different roles face different levels of transformation risk.
- Investment in education and training across career stages.
- Clear governance structures with worker representation.
- Attention to psychological and emotional impacts.
- Policy frameworks that support equitable transition.

As AORN succinctly states, "Clinicians should be full partners in preparing for the future of AI that intersects their respective practice settings" [24]. This partnership approach—treating workers as collaborators in AI adoption rather than passive recipients of technological change—offers the most promising path toward a future where AI enhances rather than diminishes the human dimensions of healthcare.

V. PROPOSED ARCHITECTURAL FRAMEWORKS FOR CLINICAL AI INTEGRATION NON-QUANT

Drawing from stakeholder submissions to the HHS RFI, this section synthesizes proposed architectural frameworks for clinical AI integration. These proposals address critical

infrastructure needs including data interoperability, governance mechanisms, and continuous monitoring capabilities, with each framework linked to relevant academic literature and established standards.

A. Shared Capacity Data Utility Model

Van Pelt & Company proposes a "shared capacity data utility" model designed as vendor-agnostic, public-private infrastructure that leverages existing systems and non-PHI operational data to provide continuously available views of healthcare system capacity. This framework addresses the foundational gap where AI tools are frequently deployed as point solutions optimized for narrow workflows but blind to system-level constraints such as staffing shortages, bed availability, and regional surge conditions [8].

The utility model operates across multiple care settings:

- **Behavioral health:** Real-time visibility into bed availability and staffing constraints enables faster placement decisions and reduces emergency department boarding. This aligns with research on AI chatbots for mental health

Figure 6: Psychological and Emotional Impacts of AI Adoption on Healthcare Workers

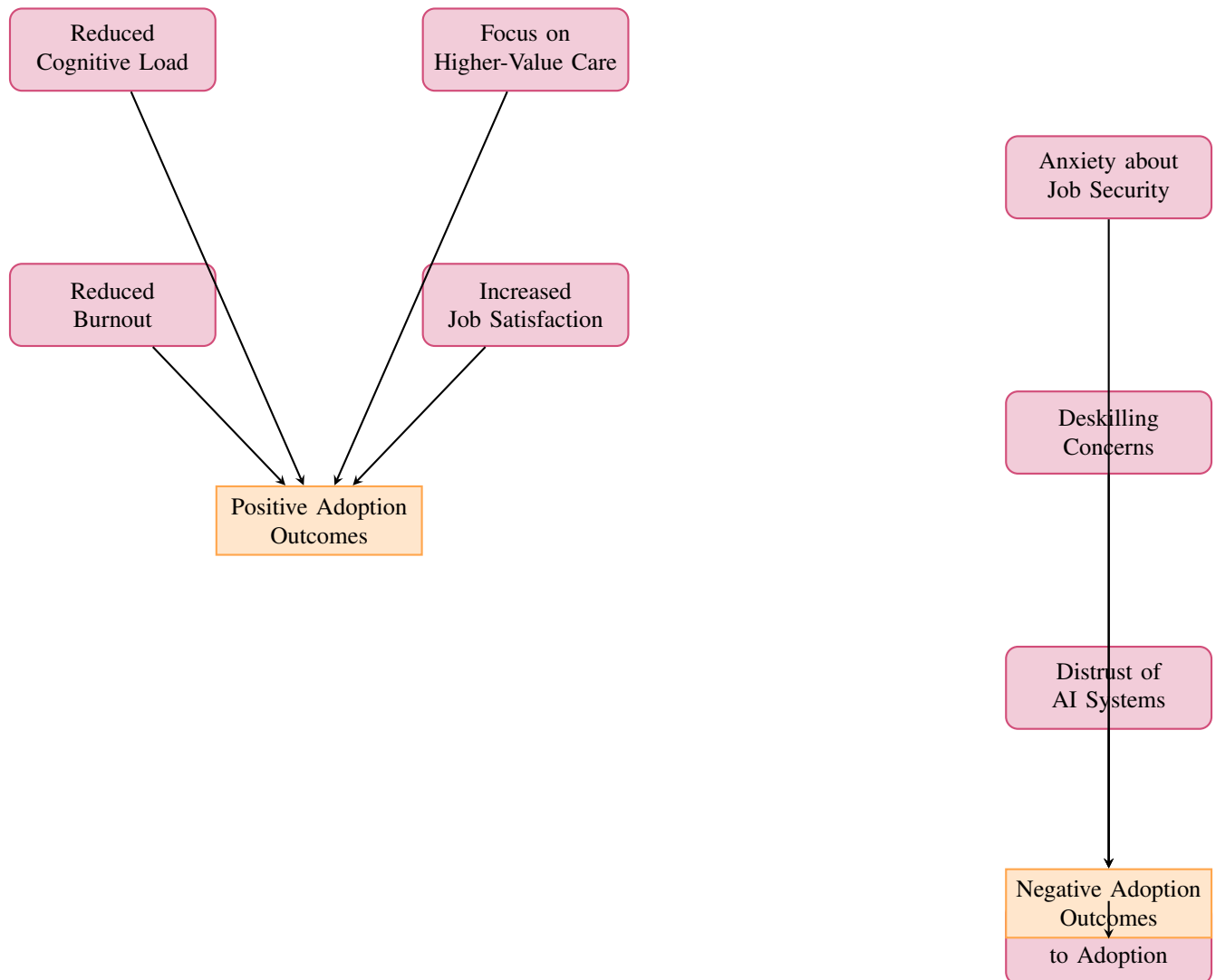


Fig. 22: Psychological and Emotional Impacts of AI Adoption on Healthcare Workers showing the dual nature of AI integration—potentially reducing burnout and cognitive load while simultaneously creating anxiety about job security and professional identity [23], [20], [11].

that emphasizes the need for contextual awareness in clinical decision support [9], [10].

- **Rural care:** Supports precise decision-making about when patients can be safely managed locally versus requiring transfer to higher-acuity care, addressing equity concerns documented in the literature on healthcare AI access disparities [11].
- **Statewide coordination:** Provides continuous situational awareness for preparedness, surge response, and alignment with federal programs, consistent with WHO guidance on

health system resilience [5].

- **Regional multi-state coordination:** Supports interoperability across jurisdictional boundaries for disaster response and medical surge scenarios, as outlined in federal interoperability frameworks [12].

This approach aligns with legislative frameworks including H.R. 2936 for real-time hospital capacity visibility and S. 1974 for healthcare system readiness coordination. The model also reflects principles from the EU AI Act regarding proportionate, risk-based oversight of AI systems [6].

B. Data Refinery Architecture for AI Readiness

Interstella's Lynqsys platform architecture emphasizes real-time data integration and AI readiness through a "Data Refinery as a Service" (DRaaS) model [13]. The proposed architecture addresses core barriers including delayed data availability, inconsistent data quality, and limited support for unstructured data. Key components include:

- **Real-time data streams:** Moving from batch processes with lag times of days or weeks to near-real-time streaming architectures that enable immediate AI analysis. This addresses data quality concerns identified in the literature where "AI systems do not fail because of algorithms; they fail because of data" [14].
- **Inline data governance:** Privacy protections and quality checks applied at the point of data creation rather than as post-hoc audits, consistent with HIPAA requirements and recent OCR guidance [15].
- **AI-ready data repositories:** Shared datasets and testbeds for AI training and evaluation, supporting the development of representative, high-quality data for model validation as emphasized in the Framework for AI Tool Assessment in Mental Health [16].
- **Support for unstructured data:** Integration of clinical notes, images, and behavioral health data, addressing the limitation that many current systems handle structured data but struggle with unstructured information critical for AI performance [2].

The architecture incorporates federated learning approaches for anonymizing data and training AI models through decentralized data repositories, enhancing accessibility while improving privacy as recommended in recent regulatory guidance [26].

C. Continuous Monitoring and Post-Deployment Surveillance Framework

Dr. Akshaya Bhagavathula proposes a "Post-Deployment Algorithmic Surveillance" (PDAS) framework for governing clinical AI as population-level interventions rather than discrete products [17]. This framework recognizes that once introduced into a health system, AI systems reshape decisions, attention allocation, workflows, and care delivery across patient populations.

The PDAS framework includes five core elements:

- 1) **Exposure characterization:** Systematic measurement of who is exposed to AI-mediated decisions, how often, and in what contexts, establishing foundational data for causal assessment.
- 2) **Continuous population-level performance monitoring:** Longitudinal evaluation of calibration, discrimination, and error patterns with mandatory stratification by key subgroups, treating performance drift as a safety signal rather than technical footnote.
- 3) **Outcome-linked evaluation:** Assessment of whether AI use improves downstream clinical outcomes, access, or utilization patterns, moving beyond surrogate metrics.

- 4) **Structured reporting of algorithmic harms:** Mechanisms to capture delayed care, silent misclassification, workflow distortions, and disparity amplification that rarely trigger traditional adverse event reports.
- 5) **Predefined regulatory response pathways:** Clear protocols for recalibration, restriction, suspension, or withdrawal when surveillance identifies concerns, with documented enforcement actions.

This framework aligns with FDA guidance on predetermined change control plans for continuously learning algorithms [7] and research on AI safety monitoring in clinical settings [4]. The emphasis on population-level effects reflects epidemiological approaches to AI governance that treat deployed AI as exhibiting properties of population-level exposures including intensity, latency, effect modification, and spillover effects [11].

D. Evidence-Based Medicine on FHIR (EBMonFHIR) Architecture

SKAF proposes an interoperable framework for representing scientific knowledge to support AI-generated clinical recommendations [18]. Building on HL7 FHIR standards, this architecture enables both human-interpretable and machine-interpretable representation of evidence supporting AI recommendations, addressing FDA guidance requiring that healthcare professionals be able to "independently review the basis for such recommendations."

The EBMonFHIR Implementation Guide defines approximately 100 profiles, extensions, and value sets for representing:

- Citations for identification, location, classification, and attribution of knowledge artifacts
- Research study design components including eligibility criteria and endpoint analysis plans
- Research results including statistic findings, variable definitions, and evidence certainty
- Aggregation and synthesis of research results across multiple studies
- Judgments regarding evidence syntheses and contextual factors related to recommendations
- Recommendations themselves with supporting evidence

A key component is the SummaryOfNetEffect Profile, which combines data elements for clinical decisions including population specification, intervention and comparator definitions, outcome considerations with risk estimates and effect changes, desirability and relative importance weighting, and net effect estimates combining contributions. This enables transparent, auditable AI recommendations that clinicians can independently evaluate, addressing concerns about generative AI producing contextually incorrect rationales to support predetermined answers [18].

The architecture supports individualized shared decision-making through tools like the NetBenefitCalculator, demonstrating how standardized evidence representation can create a "GPS for healthcare" analogous to navigation software that functions through common data sharing expectations. This approach aligns with international standards development efforts documented in the literature [29].

E. Reference Architecture for Non-Medical Device AI

BlueHalo LLC proposes a reference architecture for non-medical device AI that describes how evaluation, monitoring, interoperability, and governance components integrate across the AI lifecycle [19]. This architecture addresses the reality that many AI tools operate outside traditional regulatory pathways yet significantly influence clinical workflows.

The proposed architecture includes:

- **Evaluation layer:** Pre-deployment testing in controlled environments combined with continuous post-deployment assessment across diverse populations and settings, addressing the limitation that "validation studies answer whether an algorithm can perform under specified conditions, but they do not answer whether the system remains safe, effective, or equitable once it reshapes real-world clinical behavior" [17].
- **Monitoring layer:** Continuous drift detection, performance tracking, and safety signal surveillance with automated flagging of deviations from expected behavior, enabling early identification of degradation before adverse outcomes occur.
- **Interoperability layer:** Extended beyond data exchange to include metadata, data provenance, evaluation artifacts, and governance policies, providing context necessary for interpretation and accountability.
- **Governance layer:** Clear role definitions, oversight mechanisms, and escalation paths that reduce ambiguity and enable confident deployment in regulated environments.

This systems-level perspective recognizes that AI performance and safety depend on data quality, interoperability, human oversight, and organizational context rather than model characteristics alone [37]. The architecture supports safe harbors for organizations implementing defined governance practices, encouraging adoption of responsible AI frameworks without prescriptive mandates [38].

F. Clinical Governance and Accountability Framework

Sancian LLC proposes an operational framework for AI governance emphasizing named accountability structures and readiness assessment before deployment [20]. Key elements include:

- **Named AI Executive Owner model:** Single accountable leader with cross-functional authority to own AI outcomes, preventing the diffusion of responsibility that occurs when multiple stakeholders hold effective veto power.
- **Cross-functional AI governance councils:** Formalized structures with defined decision rights including clinical, technical, compliance, and patient safety leadership.
- **AI ReadyCheck™ assessment:** Organizational readiness baseline evaluating data governance, workflow optimization, change management capacity, and leadership alignment before technology procurement.
- **AI-EQUITYClear™ assessment:** Equity impact evaluation before deployment to identify potential disparities across demographic subgroups.

- **Decision clarity before tooling:** Governance infrastructure established before product evaluation, ensuring organizations can responsibly manage AI systems once deployed.

The framework incorporates the "AI SoftLife™ philosophy" designing AI as capacity infrastructure that reduces cognitive burden, protects clinicians' energy, and sustains humane care delivery. This human-centered approach measures success through cognitive load reduction, time returned to patient care, and clinician trust rather than technical performance alone, consistent with research on AI implementation in mental health settings [33].

G. Laboratory Data Harmonization Infrastructure

The Association for Diagnostics & Laboratory Medicine proposes infrastructure for harmonizing clinical laboratory test results to enable AI interoperability across institutions [21]. This addresses the critical barrier that different testing platforms produce varying numeric results for the same analyte, degrading model performance when data are pooled.

The proposed infrastructure includes:

- Standardized calibration protocols across assay systems
- Metadata inclusion for test methods and validation parameters in AI models
- Uniform guidelines for sample collection and reporting
- Validation requirements for AI models across diverse clinical settings
- Continuous quality monitoring including daily quality control, proficiency testing, and trend analysis

This approach ensures that AI systems using laboratory data can accurately interpret results regardless of testing site or method, supporting equitable performance across diverse healthcare settings and populations [29].

H. Dental AI Standards and Validation Framework

The Dental Standards Institute proposes a national dental AI trust framework addressing the unique challenges of AI adoption in dentistry, where diagnostic coding is inconsistently used, imaging workflows vary widely, clinical documentation remains largely unstructured, and dental and medical records remain disconnected [22].

The proposed framework includes:

- Standardized validation datasets across multiple imaging devices and clinical settings
- Third-party testing and certification mechanisms
- Minimum transparency requirements for dental AI vendors
- Standardized performance reporting formats
- Audit log requirements for AI recommendations in clinical workflows
- Guidance for post-deployment monitoring and model drift surveillance

This infrastructure supports the modernization of diagnosis capture in dentistry, enabling structured diagnostic labeling essential for AI training and validation. The framework emphasizes that "self-validation is not validation" and promotes independent benchmarking to ensure patient safety and clinical accountability.

VI. THE CURRENT LANDSCAPE OF AI IN CLINICAL CARE

Artificial intelligence has become increasingly embedded in healthcare delivery across multiple domains. As documented by the Bipartisan Policy Center, AI-enabled medical devices are a routine part of clinical practice and many patients' lives, including applications for tumor detection on imaging scans, abnormal heart rhythm identification, and digital therapeutic apps for chronic disease management [39]. The proliferation of generative AI has expanded these applications to include conversational agents for mental health support [40], [9], clinical documentation assistance, and predictive analytics for population health management [41].

However, the adoption landscape remains highly uneven. Mello and Cohen observe that health and healthcare AI tools grow in a liminal regulatory space, wedged uncomfortably between the ambits of regulatory agencies [42]. This regulatory ambiguity creates uncertainty for developers and healthcare organizations alike, potentially slowing innovation while exposing patients to unmitigated risks.

VII. REGULATORY FRAMEWORKS AND GOVERNANCE CHALLENGES

A. Federal Oversight Mechanisms

The regulatory landscape for clinical AI is fragmented across multiple agencies and frameworks. The FDA exercises jurisdiction over AI-enabled medical devices through its premarket review pathways, with recent guidance addressing predetermined change control plans for continuously learning algorithms [7], [?]. However, as De Freitas and Cohen note, most generative AI-powered wellness apps will not be reviewed by health regulators, creating a regulatory gray area where consumer-facing tools may cross into medical territory without appropriate oversight [3].

The FDA's Digital Health Advisory Committee has recently considered generative AI therapy chatbots for depression, signaling a risk-based framework for mental health tools [?]. This aligns with broader international approaches, including the EU AI Act which aims to address risks to health, safety, and fundamental rights [6].

B. State-Level Legislative Developments

State-level legislation has emerged as an additional layer of governance. Shumate et al. document that among 793 state bills reviewed, 143 were identified as potentially impactful to mental health AI, with 20 enacted across 11 states [43]. This patchwork of state approaches creates compliance complexity for developers while potentially leaving gaps in patient protections. The analysis reveals that most laws treat mental health as incidental to broader AI or healthcare regulation, with explicit mental health provisions remaining rare.

C. International Regulatory Approaches

International regulatory frameworks provide valuable models for U.S. policy development. Australia has introduced a national AI plan shaping investment, policy, and business approaches to AI adoption [44]. Canada has developed a compendium of

best practices for human-centered AI development in the world of work [45]. The UK's Medicines and Healthcare Products Regulatory Agency has responded to the Regulatory Horizons Council with guidance on AI as a medical device [?].

VIII. BARRIERS TO AI ADOPTION IN CLINICAL CARE

A. Data Readiness and Interoperability Gaps

Data fragmentation and quality issues represent primary barriers to AI adoption. The lack of harmonization among clinical data sources degrades model performance when data are pooled across institutions and limits algorithm generalizability. As Zhou and Gattinger note in their review of AI regulatory paradigms, the formation of an AI regulatory framework requires the convergence of ethical, regulatory, technical, societal, and practical considerations, with data standardization serving as a foundational element [29].

Research from Joshi's framework for government policy on agentic and generative AI in healthcare identifies data infrastructure as critical for enabling autonomous AI systems capable of real-time clinical decision support [14]. The fragmentation challenge is particularly acute in specialized domains such as dentistry, where data remains fragmented across practice management systems, imaging systems, payer workflows, and unstructured clinical notes.

B. Misaligned Incentives and Payment Barriers

Current payment models often fail to capture the value generated by AI tools that prevent adverse events or improve care coordination. In fee-for-service environments, preventing hospitalizations may reduce revenue rather than creating it, making it challenging for health systems to justify near-term investment in AI. Research from the Congressional Budget Office on AI's potential effects on the federal budget notes that current payment models may not adequately capture the long-term savings generated by preventive AI applications [?].

Nandagopal's analysis of regulatory hurdles for digital mental health technologies argues that frameworks treating AI tools as static products fail to account for their dynamic, service-oriented nature [26]. This mismatch between static product logic and dynamic service reality complicates classification, evidence generation, and post-market oversight.

C. Workforce Constraints and Change Management

Healthcare organizations operate under unprecedented workforce strain. Research from Himmelstein et al. documents that administrative work consumes one-sixth of U.S. physicians' working hours, underscoring the potential value of AI-enabled workflow optimization [31]. However, initial AI adoption requires significant change management, including workflow redesign, training and trust-building, and governance and oversight processes.

Ohu et al.'s analysis of public health risk management emphasizes that AI implementation must consider clinician cognitive load and workflow integration as critical success factors [11]. AI that adds cognitive burden rather than reducing it creates friction regardless of technical performance, and exhausted care teams cannot absorb poorly designed tools.

D. Governance Uncertainty for Non-Device AI

A significant portion of AI used in care delivery does not meet the definition of a regulated medical device yet influences clinical prioritization and care coordination. Health systems often struggle with unclear liability and accountability frameworks, uncertainty around appropriate clinical oversight, and lack of accepted frameworks for AI system governance and assessment.

Tavory’s analysis of AI regulation in mental health through an ethics of care perspective argues that existing frameworks overlook AI’s impact on human relationships and fail to establish clear duties of care toward users [33]. Golden and Aboujaoude’s Framework for AI Tool Assessment in Mental Health (FAITA-Mental Health) emphasizes the need for rigorous evaluation across credibility, user experience, crisis management, and transparency domains [16].

E. Patient Trust and Transparency Concerns

Patients consistently express concern that increasing reliance on automated systems may erode the human connection central to effective, compassionate health care. Research from Boudierhem on shaping the future of AI in healthcare through ethics and governance identifies privacy, informed consent, patient autonomy, and health equity as central ethical challenges requiring regulatory attention [46]. The WHO guidance on ethics and governance of large multi-modal models similarly emphasizes that patient trust depends on transparency, accountability, and meaningful human oversight [?].

IX. EVALUATION METHODOLOGIES FOR CLINICAL AI

A. Pre-Deployment Evaluation Frameworks

Several evaluation frameworks and reporting guidelines have been developed to facilitate healthcare organizations’ evaluation of AI-enabled technology before widespread integration. These include model cards (standardized documents explaining key characteristics of AI-enabled technology), expert consensus guidelines for reporting AI development and validation, and evaluation frameworks for healthcare professionals.

Golden and Aboujaoude’s FAITA-Mental Health framework demonstrates the application of systematic evaluation to generative AI platforms, scoring tools on credibility, user experience, crisis management, user agency, health equity, and transparency [16]. The framework effectively identifies key strengths and gaps in AI-driven mental health tools, highlighting the need for stringent standards.

B. Post-Deployment Monitoring and Continuous Evaluation

AI performance in healthcare is dynamic, with real-world safety depending not only on pre-deployment validation but on ongoing monitoring, bias detection, and clinical review. Campellone et al.’s randomized controlled trial of a generative AI digital mental health intervention provides an initial blueprint for technical and conversational guardrails that can be replicated to build safe AI systems [4].

Joshi’s comprehensive framework for AI export leadership emphasizes the importance of continuous monitoring architectures and automated compliance systems for maintaining AI safety and effectiveness over time [47]. The FDA’s guidance on predetermined change control plans enables manufacturers to update AI-enabled devices without filing additional submissions, supporting continuous improvement while maintaining safety oversight [7].

C. Human-Centered and Workflow-Based Evaluation

AI in clinical care operates within complex clinical workflows and human decision-making environments. Effective evaluation approaches include workflow impact assessments to understand how AI changes care team behavior, cognitive load and alert fatigue analysis, qualitative feedback from frontline clinicians, and evaluation of how AI supports decision-making rather than overrides it.

The guidance on responsible AI use in healthcare outlines core elements including formal AI governance committees, patient transparency, and bias monitoring for safe AI implementation [?]. This framework emphasizes that many AI failures in healthcare are not algorithmic failures but integration failures.

X. POLICY RECOMMENDATIONS

A. Modernizing Payment Models

Payment policy reform is essential to incentivize AI adoption. Recommendations include expanding CMS reimbursement pathways for AI-supported assessment, monitoring, and outreach in chronic disease management, behavioral health screening, and transitional care. Value-based payment models should recognize AI-enabled activities that demonstrably reduce total cost of care.

Coleman’s analysis of lowering healthcare costs through AI identifies value-based payment models as essential for capturing AI’s preventive benefits [?]. The Congressional Budget Office’s examination of AI’s potential effects on the economy and federal budget provides analytical support for such reforms [?].

B. Establishing Clear Governance Frameworks

Clear governance frameworks for non-device AI are needed to reduce uncertainty without over-regulating low-risk applications. Recommendations include tiered, risk-based governance pathways aligned with frameworks such as the NIST AI Risk Management Framework, with lower-risk administrative AI following streamlined pathways while higher-risk clinical decision support requires rigorous evaluation.

Joshi’s regulatory reform recommendations for agentic AI advocate for governance frameworks that distinguish between different risk levels and application contexts, integrating technical standards, risk management protocols, and policy recommendations [37]. The International Comparative Legal Guides provide overviews of current federal and state health regulatory structures for AI tools [15], [48].

C. Supporting Independent Validation and Certification

Independent validation and certification mechanisms can promote trust and accelerate adoption. Recommendations include supporting third-party audit mechanisms, standardized evaluation metrics, independent testing datasets representing diverse populations, and mandatory subgroup performance reporting.

D. Enhancing Interoperability and Data Standards

Enhanced interoperability is essential for AI development and deployment. Priority areas include integrating clinical, behavioral, and social data; enabling longitudinal, near-real-time data access; standardizing care management and engagement workflows; and supporting shared evaluation and benchmarking.

E. Patient-Centered AI Design

Patient-centered AI design must prioritize transparency, engagement, and trust. Recommendations include plain-language disclosure standards for AI use in care delivery, guidance emphasizing AI as an extension of the care team rather than replacement, and research into patient understanding and trust related to AI-supported care. The WHO guidance on AI ethics and governance emphasizes that patients should retain ownership of their personal health data with clear protections [?].

XI. RESEARCH PRIORITIES

A. Implementation Science for AI in Real-World Care Delivery

One of the largest evidence gaps is not whether AI can work but how it works when deployed at scale across diverse settings and populations. Priority research areas include AI integration into clinical workflows, change management and adoption dynamics among care teams, longitudinal performance and safety monitoring, and organizational governance models for AI oversight.

B. Equity and Bias Mitigation

Equitable AI requires evaluation beyond controlled datasets. Research priorities include performance monitoring across race, ethnicity, age, language, and socioeconomic status; stress-testing AI performance in populations with low digital literacy or limited access; and ongoing bias audits rather than one-time assessments. Joshi's framework for government policy on agentic AI emphasizes the importance of addressing algorithmic bias and ensuring equitable access [14].

C. Workforce Impact and Sustainability

AI research has historically focused on patient outcomes while under-examining workforce effects. Priority areas include impact of AI on clinician workload and burnout, redistribution of tasks across care teams, and training and competency development for AI-supported care. Analysis notes that AI adoption will not scale if it increases burden or undermines workforce sustainability [49].

D. Cost-Benefit Analysis and Economic Impact

The literature on AI cost impact remains fragmented and inconsistent. Research priorities include total cost of care impact over time, attribution models for AI-enabled prevention, financial effects across different payment models, and labor costs for equivalent tasks performed by AI. Clear evidence of cost reduction is essential to align incentives and accelerate adoption.

DECLARATION

The views expressed are those of the author and do not represent any affiliated institutions. This work is conducted as part of independent research. This is a review paper, and all results, proposals, and findings are derived from the cited literature. The author does not claim any novel findings. The author's work was to review and organize existing research.

Portions of this manuscript were drafted with the assistance of AI writing tools (including ChatGPT/Claude) to improve clarity and organization. All AI-generated content was reviewed, edited, and verified by the author for coherence, and to eliminate potential hallucinations as much as possible. The LaTeX code was developed with the assistance of GitHub Copilot and edited through DeepSeek. Final responsibility for all content, including any errors or omissions, rests solely with the readers. This is a working paper and edits are expected in the next version.

XII. CONCLUSION

Artificial intelligence has reached a point of maturity where it can meaningfully improve patient safety, strengthen care delivery, and reduce the total cost of care—but only if deployed with empathy, accountability, and strong clinical governance. This comprehensive analysis of stakeholder submissions to the HHS RFI reveals that the greatest opportunity for AI in clinical care lies not in automation for its own sake, but in augmenting human care teams, engaging patients earlier and more consistently, and preventing avoidable deterioration before it becomes costly or harmful.

A. Key Findings

Our synthesis of 15 stakeholder submissions and supporting literature yields several critical insights:

First, the primary barriers to AI adoption are not technical but structural: data interoperability gaps, fragmented governance across federal and state agencies, misaligned reimbursement incentives that reward volume over prevention, and workforce constraints that leave healthcare organizations with insufficient capacity to manage change. As multiple stakeholders emphasized, "AI systems do not fail because of algorithms; they fail because of data."

Second, stakeholder-proposed architectural frameworks offer a coherent roadmap for addressing these barriers. The shared capacity data utility model provides vendor-agnostic infrastructure for real-time operational visibility across behavioral health, rural care, and emergency response. The PDAS framework shifts evaluation from static pre-market approval to continuous

population-level monitoring, treating deployed AI as dynamic interventions requiring ongoing surveillance. The EBMonFHIR architecture enables transparent, auditable AI recommendations through standardized representation of clinical evidence.

Third, successful AI governance requires named accountability structures. Sancian LLC's AI Executive Owner model and cross-functional governance councils address the diffusion of responsibility that currently slows adoption. These frameworks recognize that AI safety depends as much on organizational readiness as on technical performance.

Fourth, the future of work with AI adoption presents both challenges and opportunities. While routine pattern-recognition roles face automation pressure, the predominant stakeholder perspective emphasizes augmentation over replacement. AI can reduce cognitive burden, alleviate burnout, and enable clinicians to focus on higher-value patient interactions—but only if workers are treated as partners in transformation rather than passive recipients of technological change. Emerging roles—AI Clinical Governance Officer, Clinical AI Specialist, AI Ethics & Equity Officer—require new educational competencies spanning technical literacy, critical evaluation, ethical awareness, and workflow integration.

B. Patient Safety as Foundation

Patient safety must remain the cornerstone of AI deployment in healthcare. The most effective AI systems operate within clearly defined clinical guardrails, incorporate human oversight, and are continuously monitored in real-world use. Engagement and empathy are safety features—AI tools that patients trust and interact with consistently are more likely to detect emerging concerns and less likely to miss critical signals. As Ellipsis Health emphasizes, empathy is not merely a usability feature but a patient safety imperative.

C. Policy Imperatives

HHS has a unique and critical role to play in accelerating safe and effective AI adoption. Federal leadership should:

- **Align reimbursement with prevention and engagement**, expanding CMS pathways for AI-supported assessment, monitoring, and outreach while ensuring that value-based payment models recognize AI-enabled activities that demonstrably reduce total cost of care.
- **Clarify governance expectations for non-device AI**, issuing guidance on appropriate oversight models, human-in-the-loop review, and documentation standards proportional to risk. Tiered, risk-based pathways aligned with frameworks such as the NIST AI Risk Management Framework can reduce uncertainty without over-regulating low-risk applications.
- **Support standardized evaluation and accreditation frameworks**, including third-party audit mechanisms, independent testing datasets representing diverse populations, and mandatory subgroup performance reporting. As the Dental Standards Institute notes, "self-validation is not validation."

- **Reduce administrative friction and duplication** through mutual recognition of credible private-sector accreditation programs and portability of evaluations across federal and state programs.
- **Invest in real-world implementation science** and public-private partnerships that generate evidence on AI integration, workforce impact, and long-term outcomes. The PDAS framework exemplifies the shift from episodic validation to lifecycle-based oversight.
- **Fund workforce development programs** targeting AI-related skills across career stages, from foundational education through continuing professional development. Educational institutions must integrate AI competencies into curricula, emphasizing ethical reasoning alongside technical proficiency.

D. A Call to Action

AI represents a rare opportunity to advance the shared goals of patient safety, better outcomes, workforce sustainability, and cost containment. Realizing this opportunity requires:

- **Patient-centered design** grounded in empathy and trust, with plain-language disclosure standards that explain AI involvement to patients and preserve the human connection central to compassionate care.
- **Strong clinical guardrails and continuous quality assurance**, including input guardrails for crisis detection, model guardrails constraining output generation, output guardrails validating against clinical guidelines, and escalation guardrails triggering human involvement when safety thresholds are exceeded.
- **Incentives that reward prevention rather than volume**, modernizing payment models to capture the long-term savings generated by early detection, proactive engagement, and avoided utilization.
- **Federal policies that reduce uncertainty and accelerate adoption**, providing clear rules of the road while enabling innovation through regulatory sandboxes, safe harbors for defined governance practices, and coordinated action across HHS operating divisions.

E. Final Reflection

As the Association of perioperative Registered Nurses succinctly states, "Clinicians should be full partners in preparing for the future of AI that intersects their respective practice settings." This partnership approach—treating workers, patients, and communities as collaborators in AI adoption rather than passive recipients of technological change—offers the most promising path toward a future where AI enhances rather than diminishes the human dimensions of healthcare.

The architectural frameworks, governance structures, and policy recommendations synthesized in this paper provide a roadmap for that future. By treating AI as care infrastructure requiring coordinated investment across data, standards, monitoring, and governance—rather than isolated technology deployment—HHS can accelerate adoption while maintaining patient safety and public trust. The next 24-36 months represent

a narrow window to embed these principles before AI becomes irreversible infrastructure. With thoughtful leadership and collaboration, AI can help make healthcare safer, more humane, and more affordable for all Americans.

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